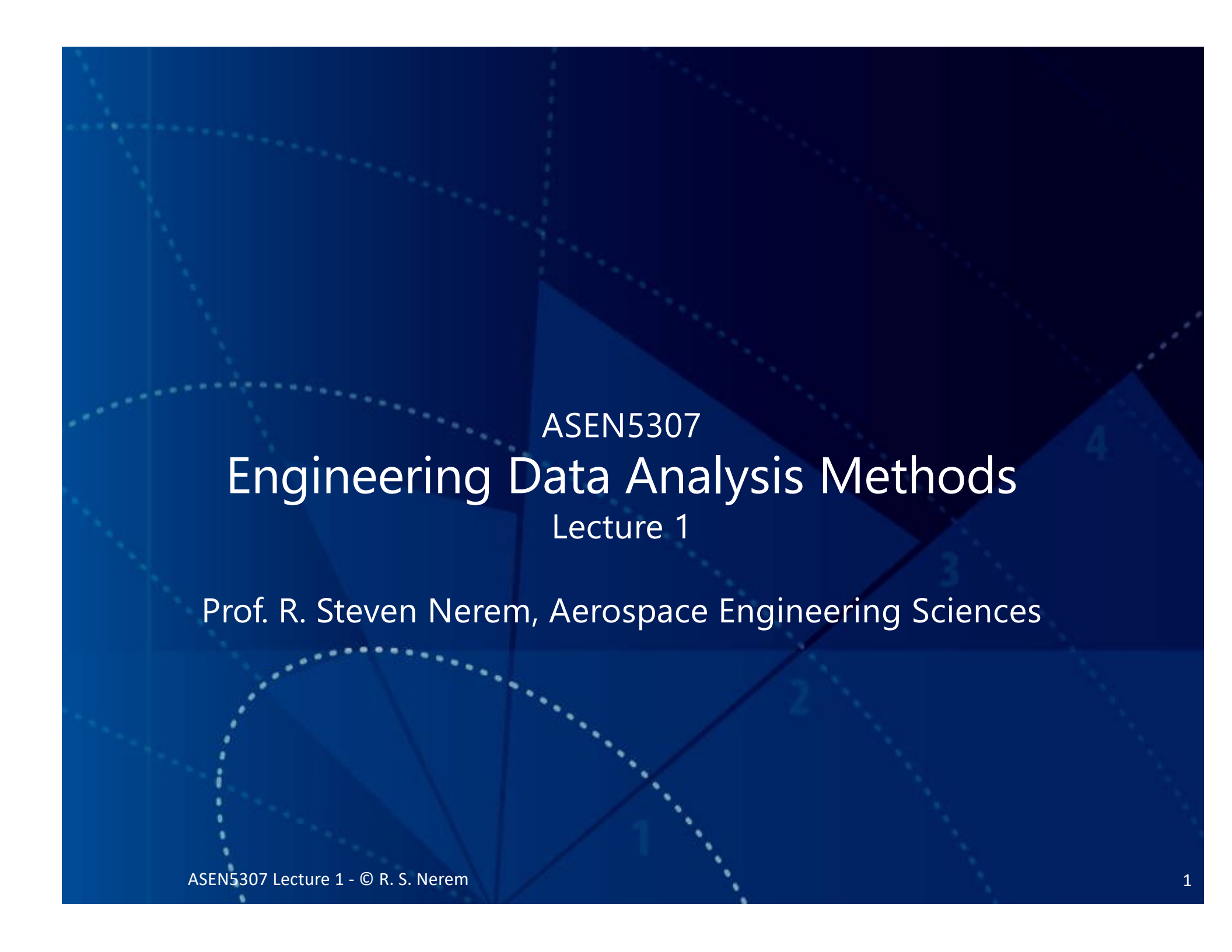




ASEN5307

Engineering Data Analysis Methods

Lecture 1

Prof. R. Steven Nerem, Aerospace Engineering Sciences

ASEN5307 – Engineering Data Analysis

I. Introduction

1. Collecting Data
2. Data Calibration and Interpolation
3. Data Editing
4. Presenting Data

II. Statistics and Error Handling

1. Uncertainties in Measurements
2. Empirical Distributions
3. Theoretical Distributions
4. Hypothesis Testing
5. t-test, F-test, χ^2 test
6. Error Analysis
7. Error Propagation
8. Confidence Intervals
9. Correlation Coefficient
10. Degrees of Freedom
11. Estimation Methods
12. Curve Fitting
13. Interpolating Data
14. Data Smoothing (boxcar, gaussian, Savitzky–Golay)
15. Covariance and Error Analysis
16. Residual Analysis and Data Editing
17. Linear Regression Analysis
18. Bootstrap and Jackknife Estimates

ASEN5307 – Engineering Data Analysis

III. Time-Series Analysis

1. Fourier Analysis
2. Harmonic Analysis
3. Blackman-Tukey Spectral Analysis
4. Cross-Spectral Analysis
5. Wavelet Analysis
6. Analyzing Unevenly Spaced Data
7. Lomb-Scargle Powerspectrum

IV. Signal Processing

1. Linear Time Invariant Systems
2. Convolution and Filtering
3. Recursive and Nonrecursive Filters
4. Impulse and Frequency Response
5. Filter Design

V. Spatial Analysis of Data Fields

1. Gridding and Contouring
2. Spherical Harmonics
4. Objective Analysis, Kriging
5. Principal Component Analysis (including SVD)
6. Independent Component Analysis
7. Empirical Orthogonal Functions

VI. Machine Learning

ASEN5307 – Engineering Data Analysis

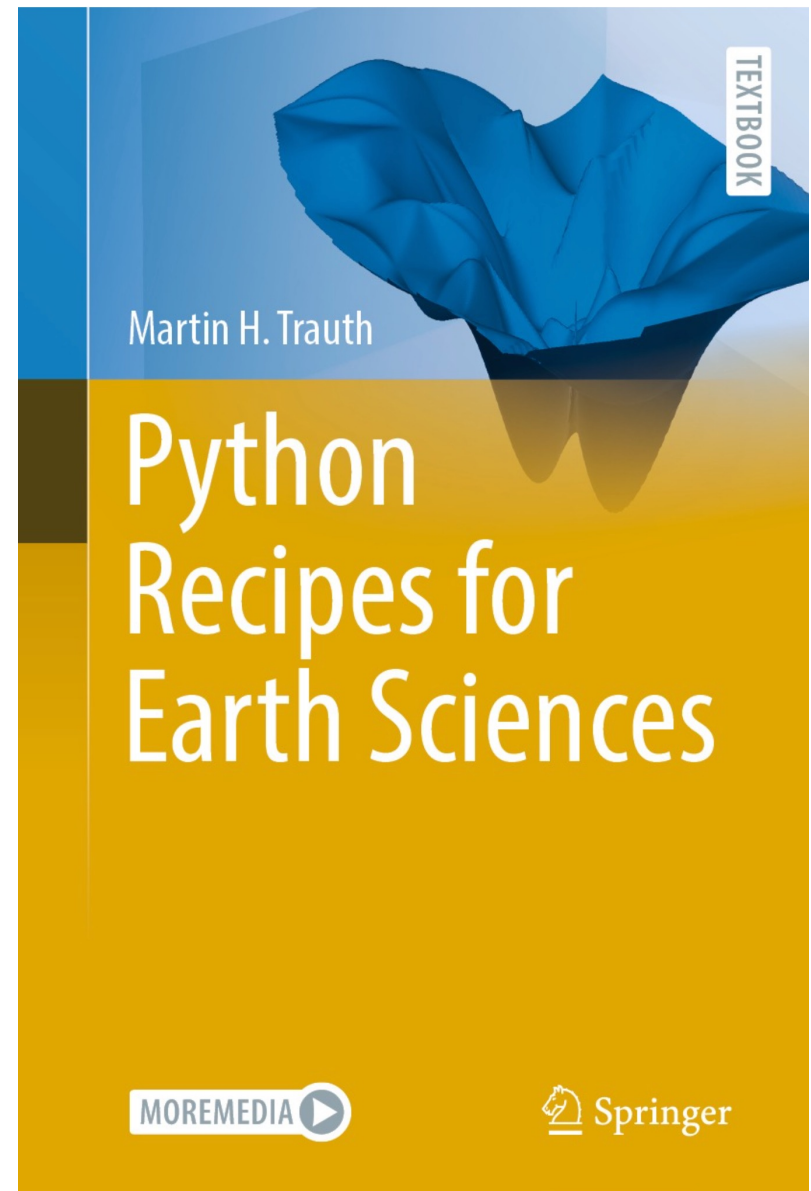
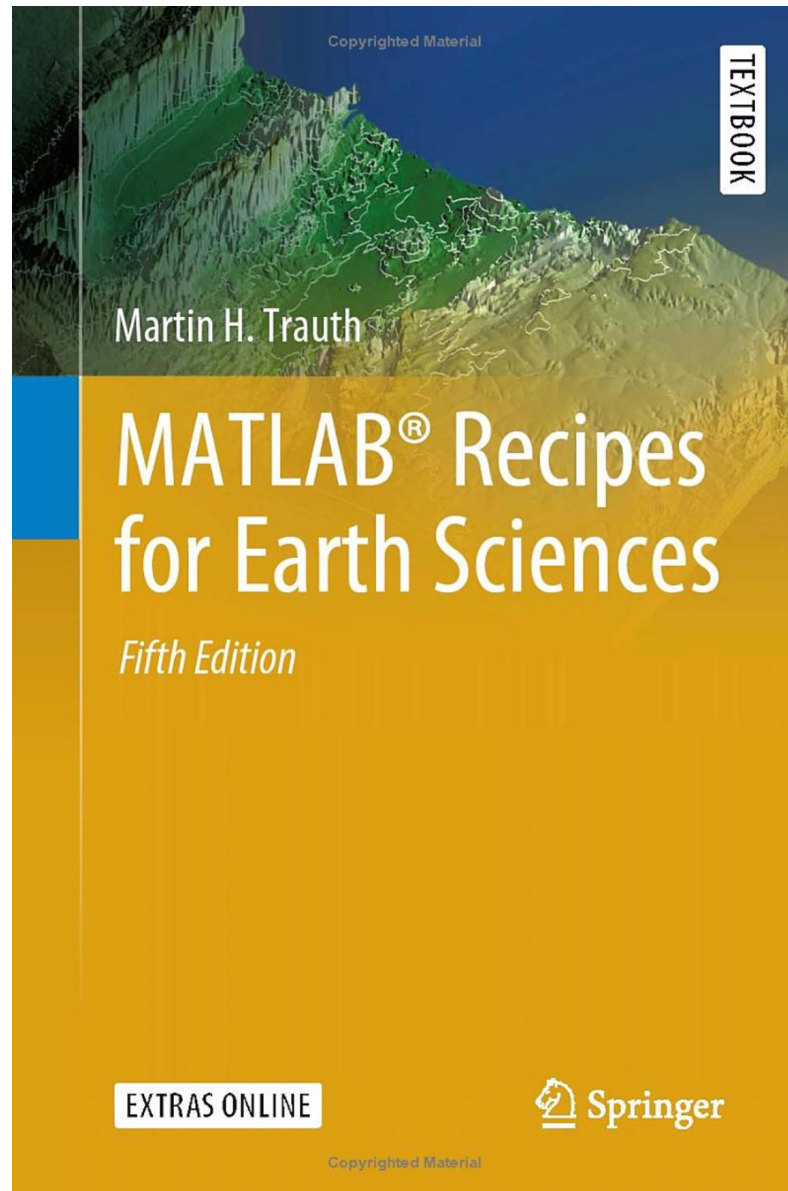
Instructor	Dr. R. Steven Nerem (Office: AERO 456, Ph. 492-6721, Email: nerem@colorado.edu)
Class Time	TTh 1 – 2:15 pm
Class Location	AERO 111
Class Web Page	http://canvas.colorado.edu
Office Hours	9-10 am Tuesdays and Thursdays or by appointment
Teaching Assistant	Lluc Busquets (lluc.busquets@colorado.edu)
Required Text	<i>MATLAB Recipes for Earth Sciences</i>, 2021, 5th Edition by Martin H. Trauth, Springer ISBN-13: 978-3030384401 or <i>Python Recipes for Earth Sciences</i>, 2022, 1st Edition by Martin H. Trauth, Springer ISBN-13: 978-3031077180
Grading	Mid-Term Exam (20%), Final Exam (20%) Homework (40%) (10 pts deducted for each day late!) Research Project (20%)
Important Dates	October 8 – Reading Day (no class) October 13 – In-Class and Take-Home Mid-Term Exam December 3 – In-Class and Take-Home Final Exam December 3 – Research Projects Due (Canvas Upload) December 8 – 4:30 pm – Research Project Presentations
Lecture Material	PDF files will be posted on the class website.
Course Overview	Gives students broad exposure to a variety of traditional and modern statistical methods for filtering and analyzing data. Topics include estimation and filtering methods, principal component analyses and spectral analyses. Introduces these methods and provides practical experience with their use. Students carry out problem assignments. <u>Prior knowledge of MATLAB and/or Python is recommended, as programming is NOT taught in this class.</u>

ASEN 5307 Fall 2026 Schedule

Week	Date	Topics	Reading	HW	Exams
1	Tue, Aug 18, 2026	No Class			
Aug 18, 2026	Thu, Aug 20, 2026	L1: Course Overview, Error Analysis	Ch. 3		
2	Tue, Aug 25, 2026	L2: Descriptive Statistics	Ch. 3	HW 1 Assigned	
Aug 25, 2026	Thu, Aug 27, 2026	L3: Theoretical Distributions	Ch. 3		
3	Tue, Sep 1, 2026	L4: Theoretical Distributions	Ch. 3	HW 2 Assigned HW 1 Due	
Sep 1, 2026	Thu, Sep 3, 2026	L5: Robust Statistics	Ch. 3		
4	Tue, Sep 8, 2026	L6: Curve Fitting	Ch. 4	HW 3 Assigned HW 2 Due	
Sep 8, 2026	Thu, Sep 10, 2026	L7: Curve Fitting	Ch. 4		
5	Tue, Sep 15, 2026	L8: Correlation and Regression	Ch. 4	HW 4 Assigned HW 3 Due	
Sep 15, 2026	Thu, Sep 17, 2026	L9: More Curve Fitting	Ch. 4		
6	Tue, Sep 22, 2026	L10: Introduction to Time Series Analysis	Ch. 5	HW 5 Assigned HW 4 Due	
Sep 22, 2026	Thu, Sep 24, 2026	L11: Fourier Analysis	Ch. 5, Notes		
7	Tue, Sep 29, 2026	L12: Fourier Analysis	Ch. 5, Notes	HW 6 Assigned HW 5 Due	
Sep 29, 2026	Thu, Oct 1, 2026	L13: Spectral Analysis	Ch. 5, Notes		
8	Tue, Oct 6, 2026	L14: Spectral Analysis	Ch. 5, Notes	HW 6 Due	
Oct 6, 2026	Thu, Oct 8, 2026	Reading Day - No Class			
9	Tue, Oct 13, 2026	Midterm Exam			Exam
Oct 13, 2026	Thu, Oct 15, 2026	L15: Digital Filtering	Ch. 6		
10	Tue, Oct 20, 2026	L16: Digital Filtering	Ch. 6	HW 7 Assigned	
Oct 20, 2026	Thu, Oct 22, 2026	L17: Digital Filtering	Ch. 6		
11	Tue, Oct 27, 2026	L18: PCA/EOFs	Ch. 9	HW 7 Due	
Oct 27, 2026	Thu, Oct 29, 2026	L19: PCA/EOFs	Ch. 9	HW 8 Assigned	
12	Tue, Nov 3, 2026	L20: EOFs	Ch. 9		
Nov 3, 2026	Thu, Nov 5, 2026	L21: PCA and Clustering	Ch. 9	HW 8 Due	
13	Tue, Nov 10, 2026	L22: Introduction to Machine Learning	Notes		
Nov 10, 2026	Thu, Nov 12, 2026	L23: Machine Learning	Notes	HW 9 Assigned	
14	Tue, Nov 17, 2026	L24: Machine Learning	Notes		
Nov 17, 2026	Thu, Nov 19, 2026	L25: Machine Learning	Notes		
15	Tue, Nov 24, 2026	Thanksgiving Break			
Nov 24, 2026	Thu, Nov 26, 2026	Thanksgiving Break			
16	Tue, Dec 1, 2026	L26: Spherical Harmonics	Notes	HW 9 Due	
Dec 1, 2026	Thu, Dec 3, 2026	Final Exam			Exam
	Finals Week	Project Presentations			

Note: schedule is subject to change throughout the semester

Textbooks



Programming

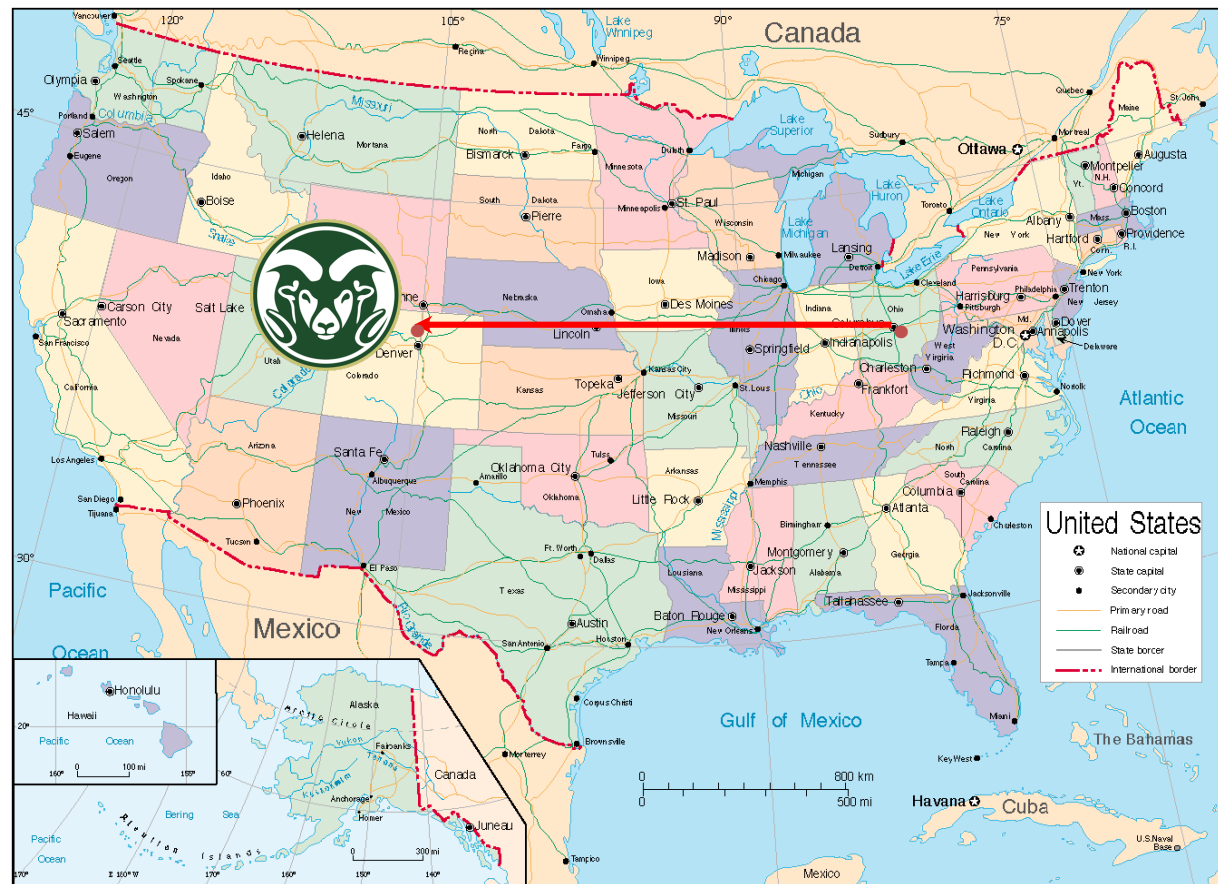
- You can use either Matlab or Python in this class, however, data files used in homework assignments will be given in Matlab or text format (which can both be easily read with Python).
- **This is not a programming class.** The TA and I will not be able to help you with finding bugs in your code. You are expected to do this on your own.
- Therefore, some background with Matlab or Python is recommended, or the commitment to learn one of these languages during the semester.

Nerem's Travels



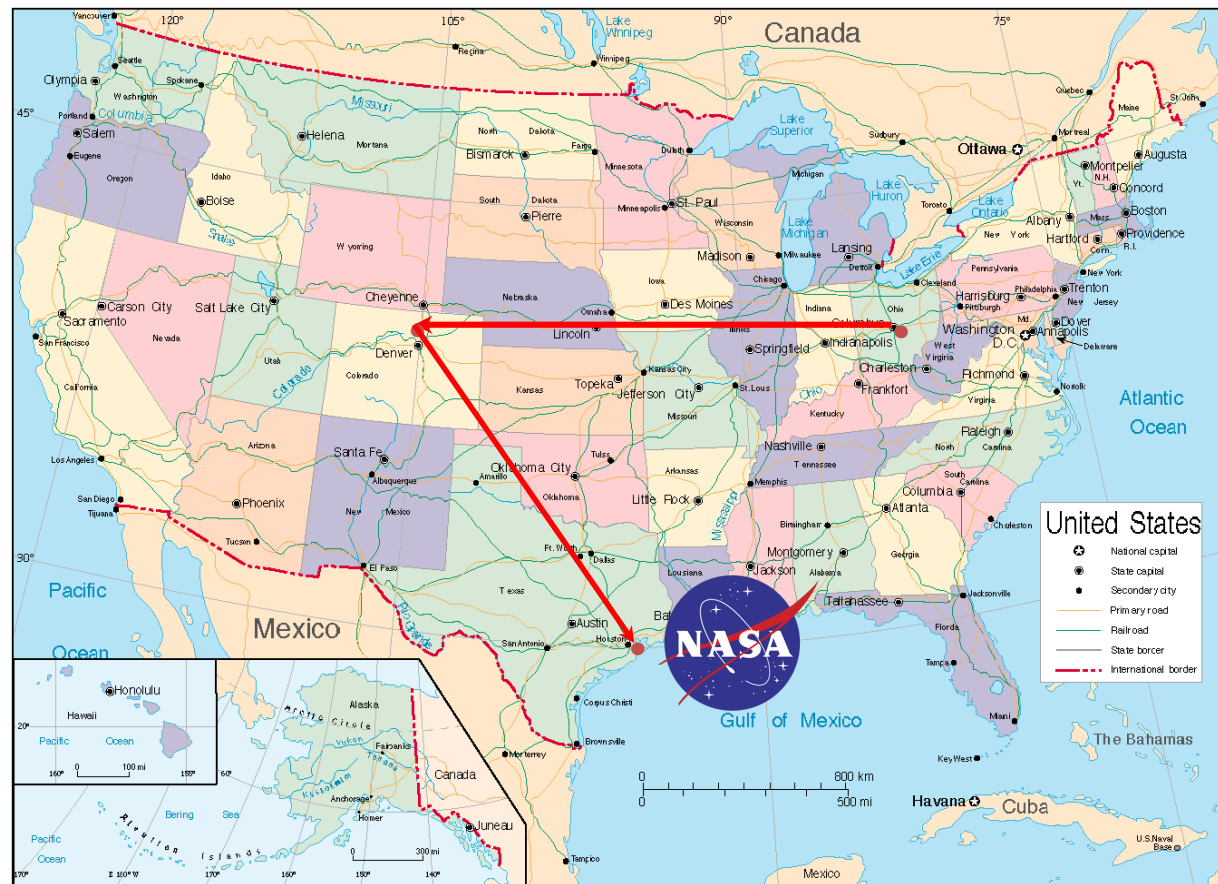
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Nerem's Travels



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Nerem's Travels

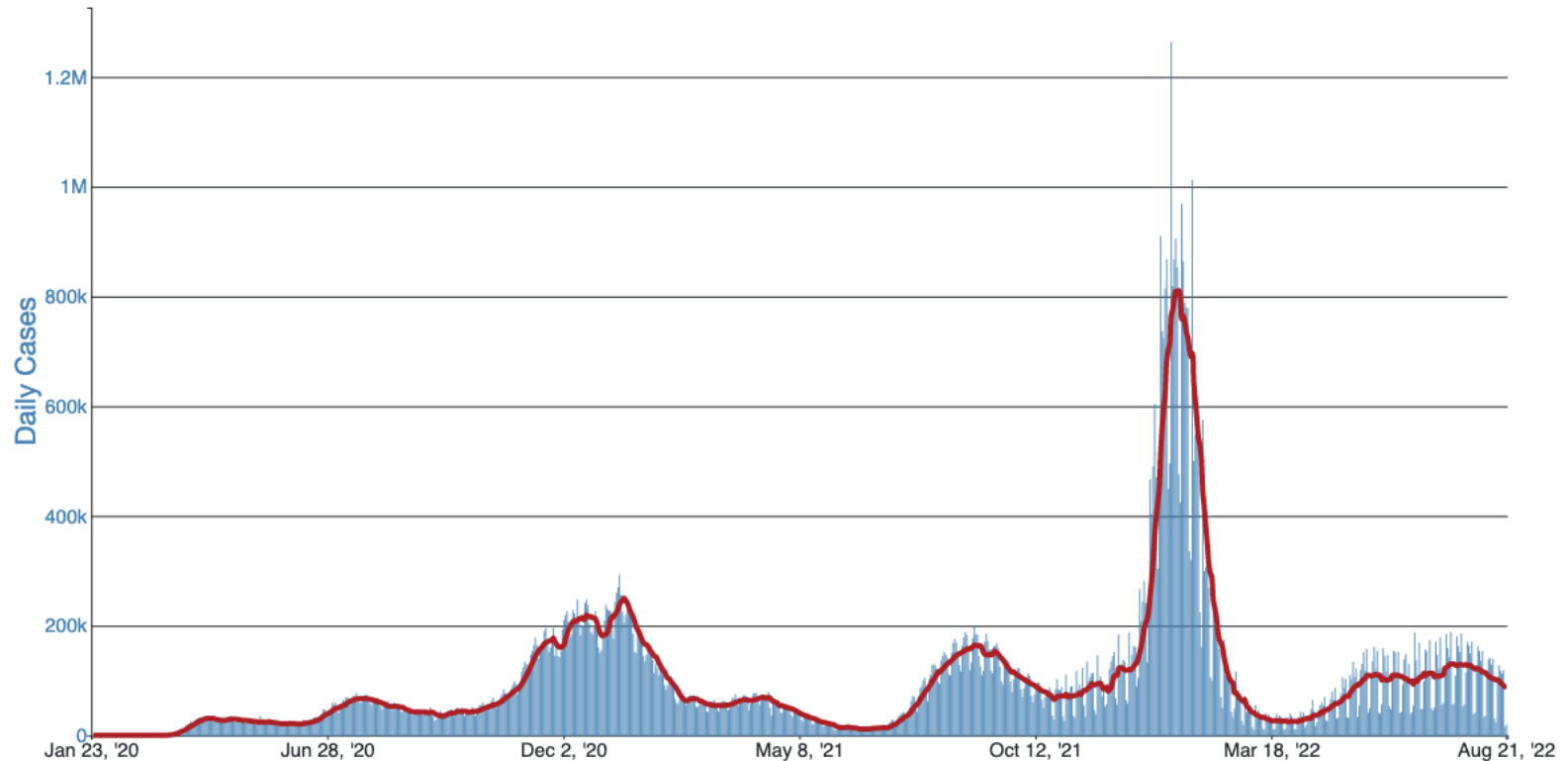


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Types of Data

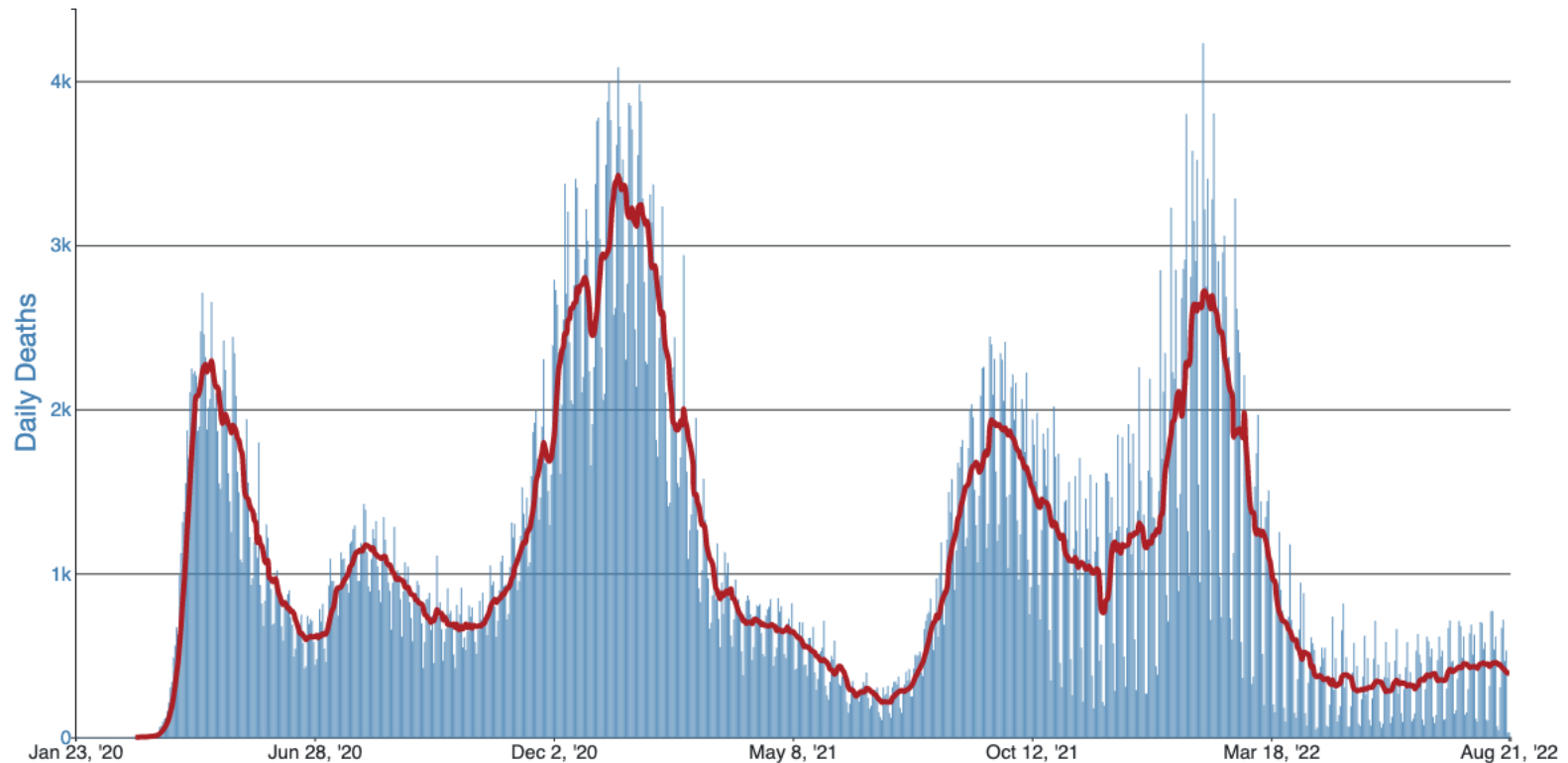
COVID Data

Daily Trends in Number of COVID-19 Cases in The United States Reported to CDC



COVID Data

Daily Trends in Number of COVID-19 Deaths in The United States Reported to CDC



COVID Data

United States | All Ages

5,181,230

Total Admissions

Aug 01, 2020 - Aug 20, 2022

5,047

Current 7-Day Average

Aug 14, 2022 - Aug 20, 2022

5,892

Prior 7-Day Average

Aug 07, 2022 - Aug 13, 2022

21,525

Peak 7-Day Average

Jan 09, 2022 - Jan 15, 2022

-14.3%

Percent change from prior 7-day
avg. of Aug 07, 2022 - Aug 13, 2022

-76.6%

Percent change from peak 7-day
avg. of Jan 09, 2022 - Jan 15, 2022

New Admissions of Patients with Confirmed COVID-19, United States

Aug 01, 2020 - Aug 20, 2022

By Jurisdiction and Age Group

By Jurisdiction

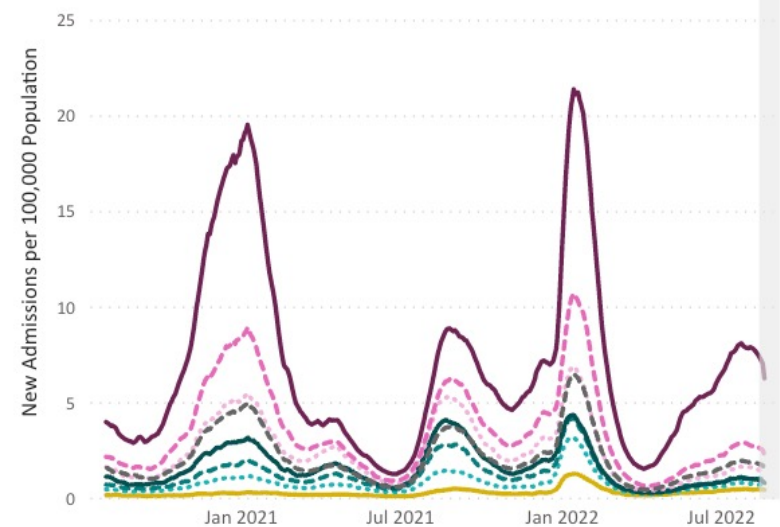
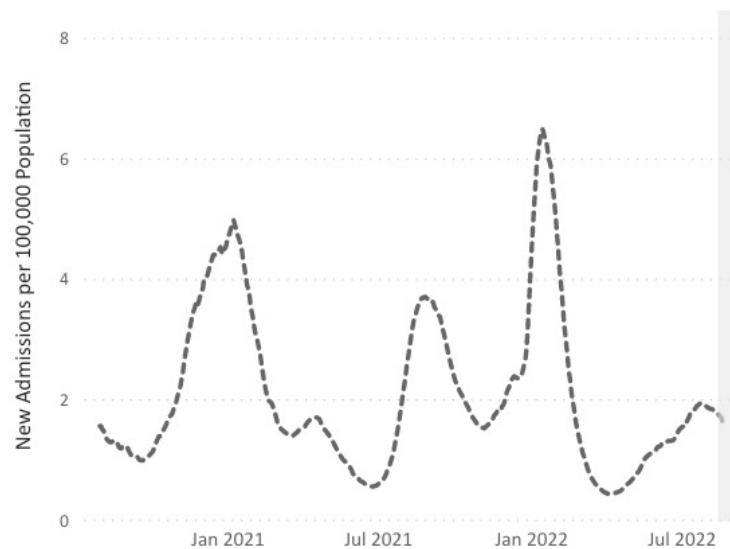
Select a Jurisdiction

United States

Select an Age Group

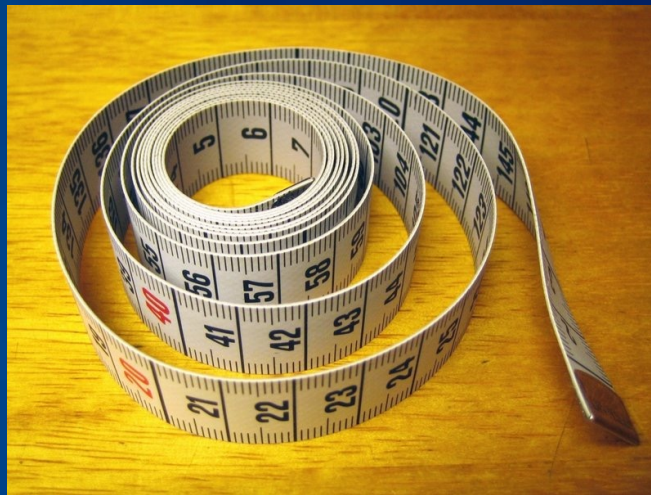
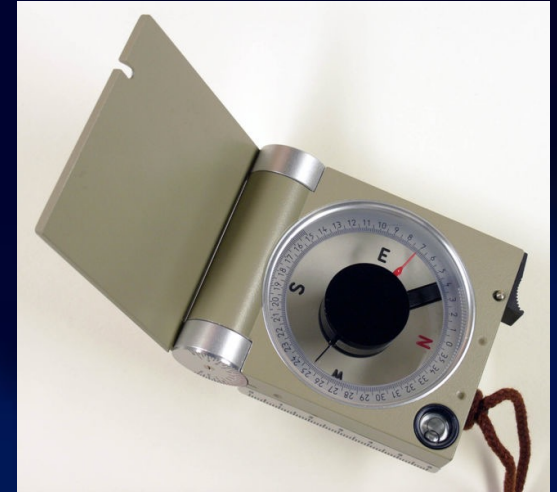
All Ages

United States | All Ages



Age Group — 0 - 17 Years — 18 - 29 Years — 30 - 39 Years — 40 - 49 Years — 50 - 59 Years — 60 - 69 Years — 70+ Years — — All Ages

Types of Data



(Wikipedia Commons, GNU, Leica Systems and Department of Geosciences, University of Potsdam)
ASEN5307 Lecture 1 - © R. S. Nerem

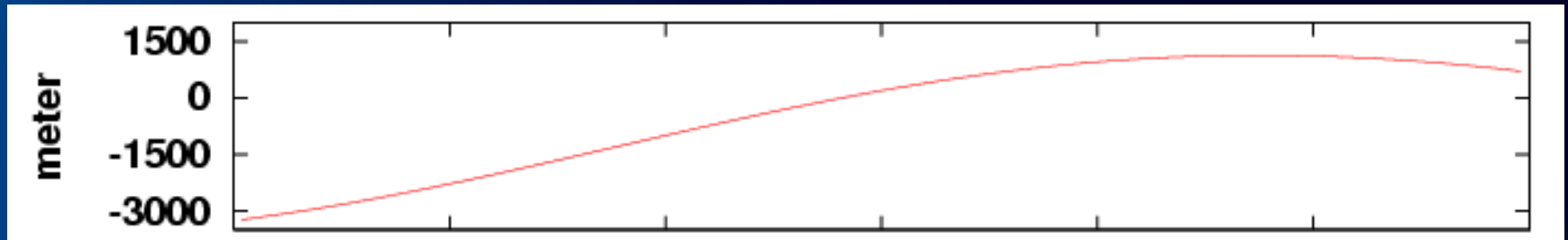
Types of Data

A	B	C	D	E	F	G
1	630	319	26.1	35.4	1	1
2	630	319	27.7	36.8	2	1
3	621	315	30.3	40.9	1	1
4	621	315	30.9	43.4	2	1
5	625	332	37.1	58.4	1	1
6	625	332	39.0	61.1	2	1
7	642	322	23.3	38.6	1	2
8	642	322	24.2	36.0	2	2
9	633	307	30.4	44.0	1	2
10	633	307	*	*	2	2
11	611	324	35.3	53.6	1	2
12	611	324	*	*	2	2
13	619	339	37.5	52.4	1	2
14	619	339	38.9	53.9	2	2
15	618	310	21.8	38.6	1	2
16	618	310	23.2	38.1	2	2

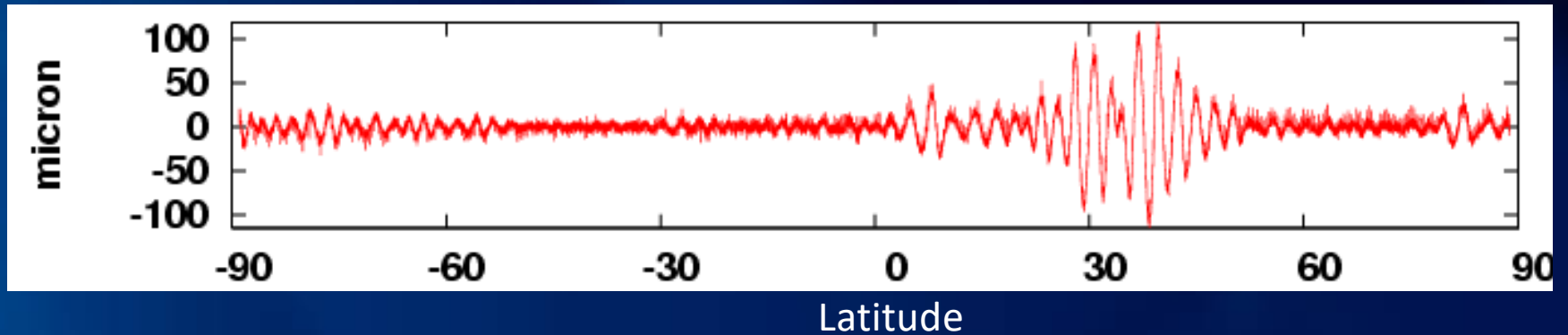
(Swan and Sandilands, 1996, Page 2, Box 1.2)

Types of Data

(a)

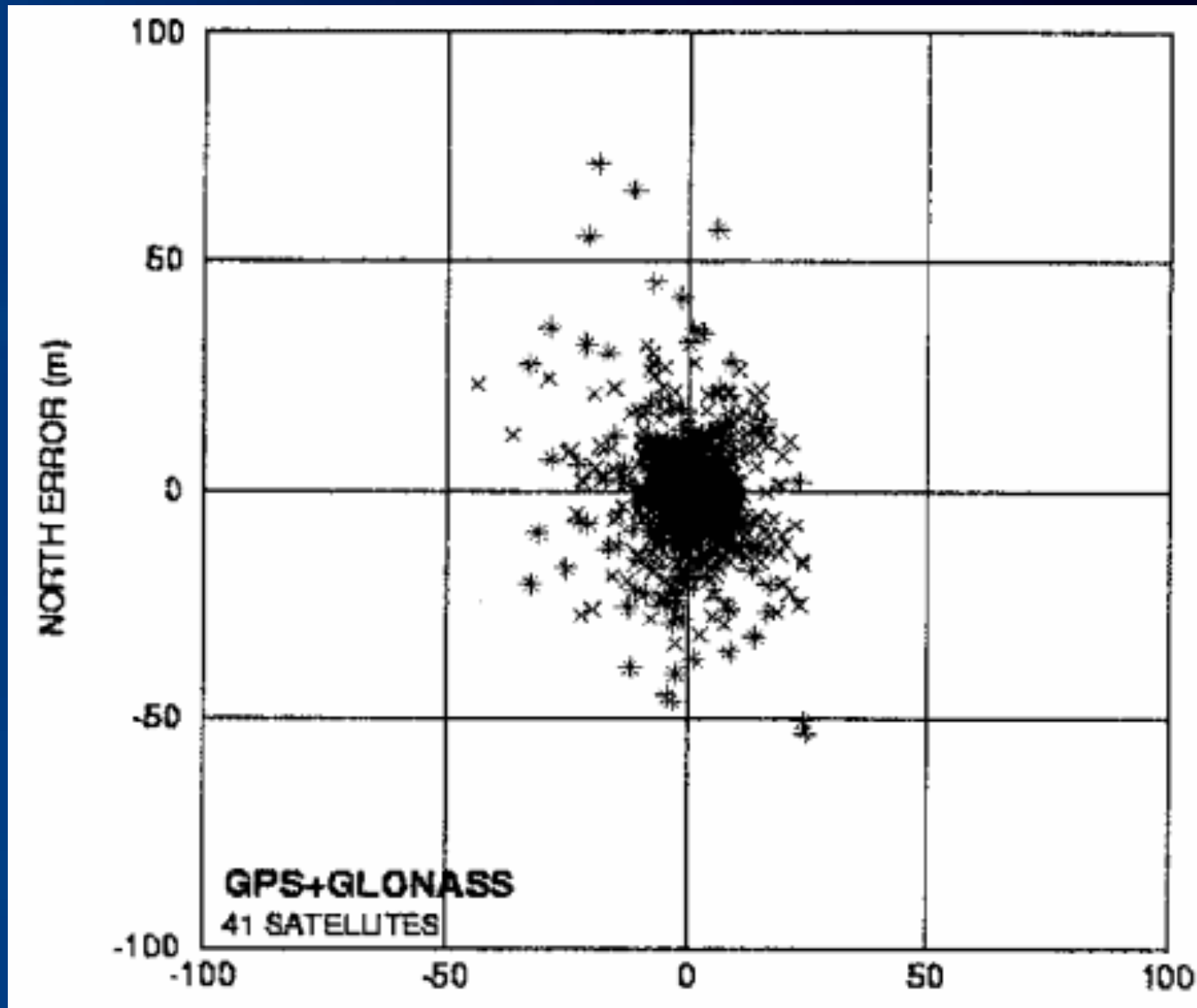


(b)

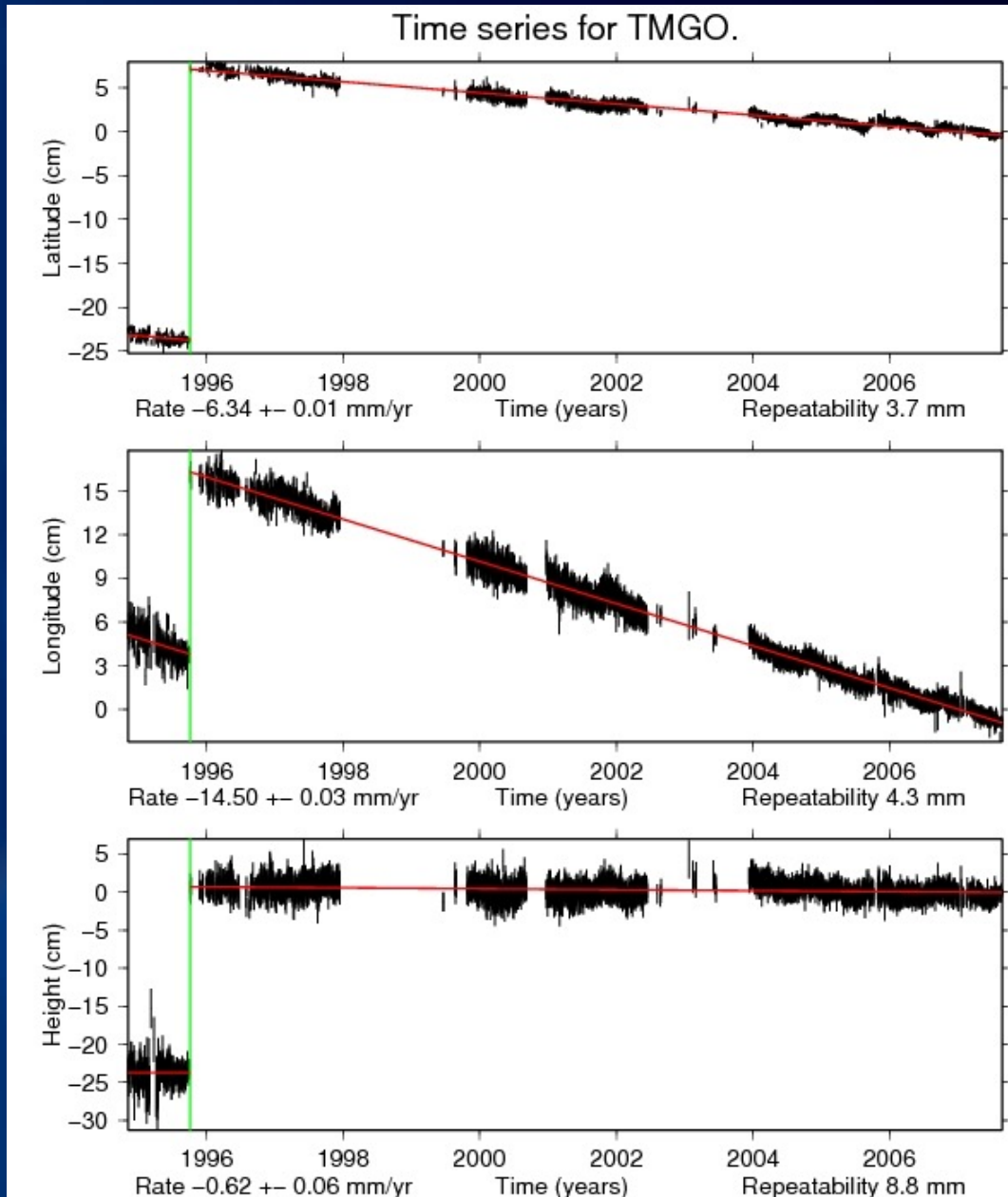


(a) Range between the GRACE satellites and (b) after high-pass filtering

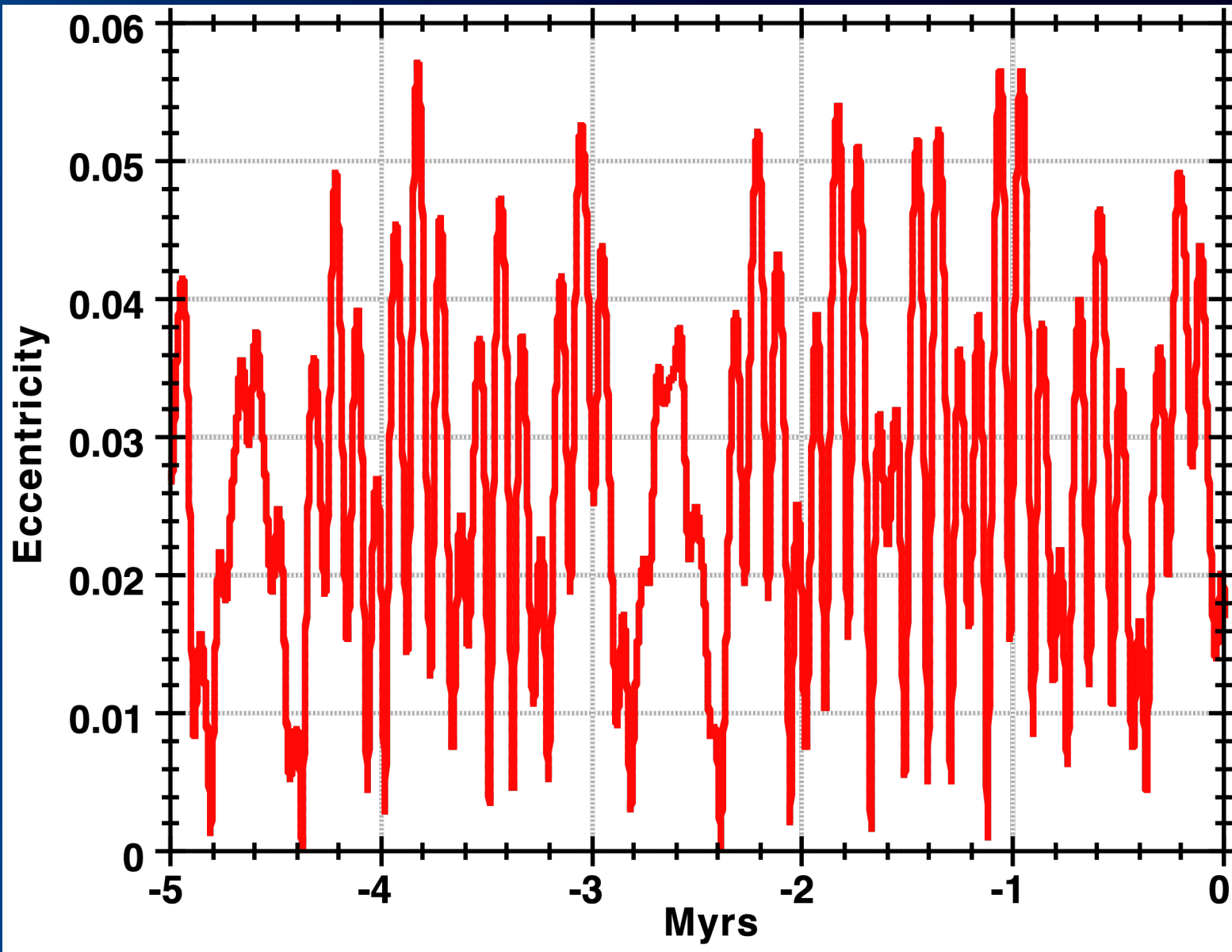
Types of Data: GPS Positions



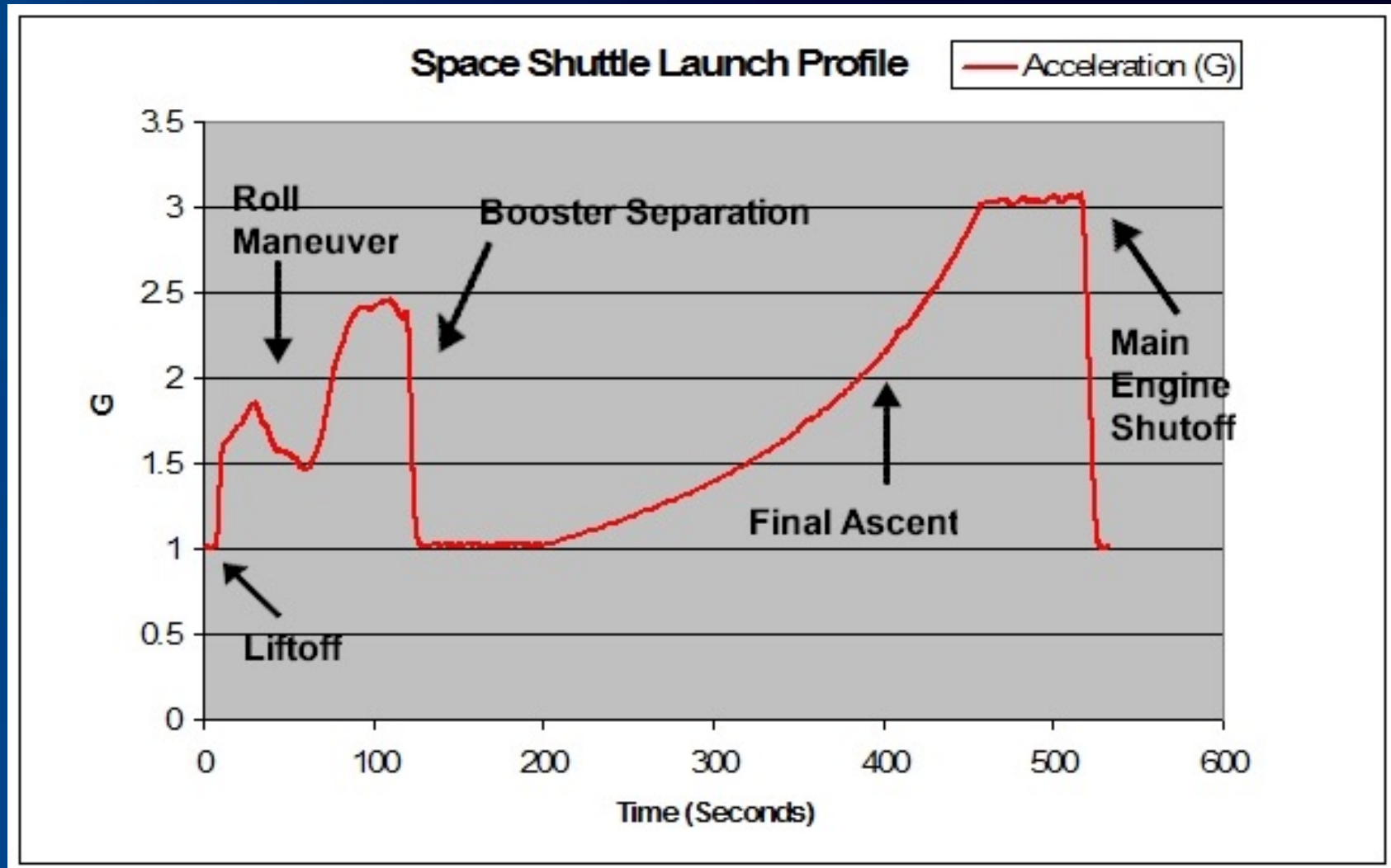
Types of Data: GPS Positions



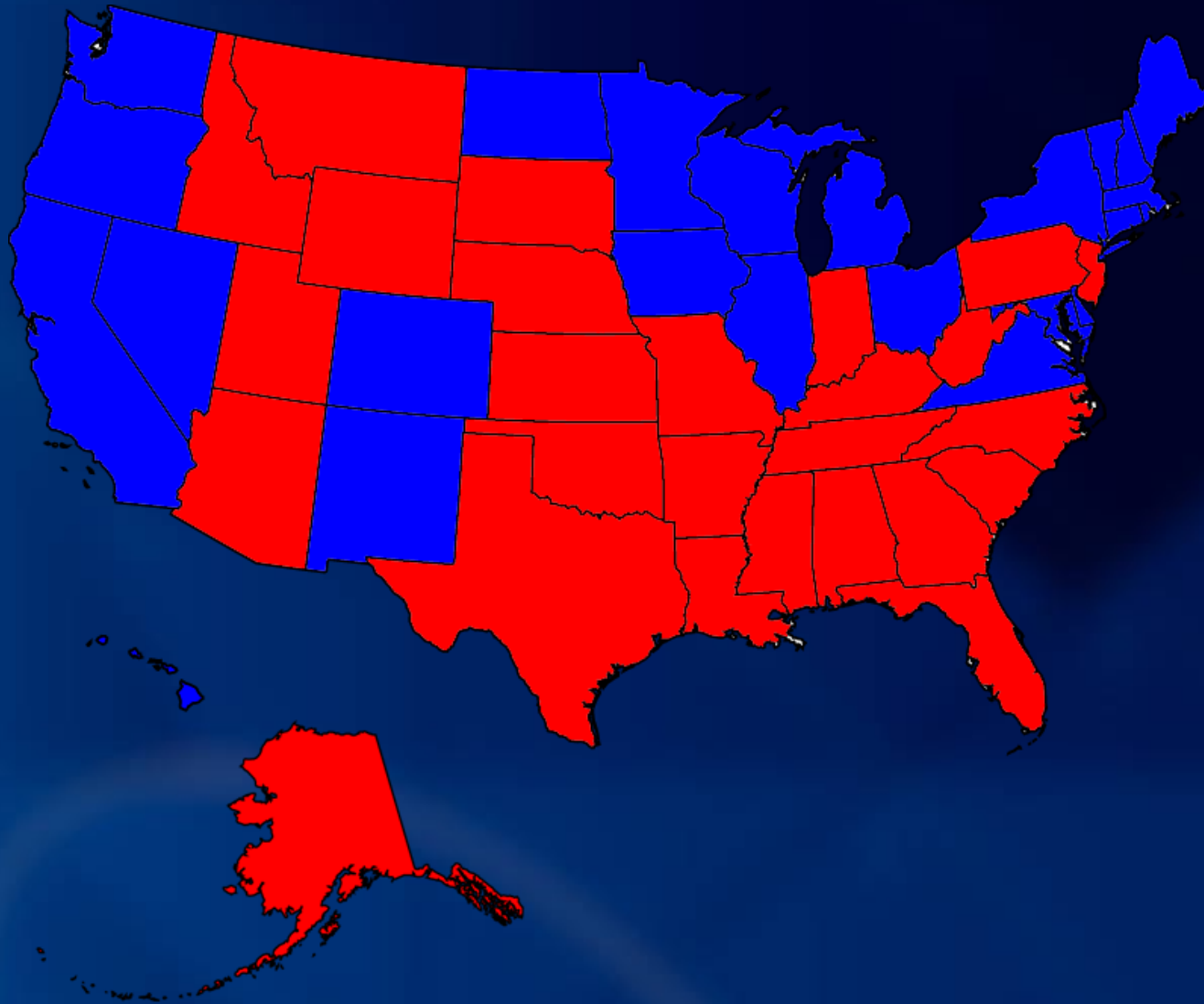
Types of Data: Earth's Orbit Variations



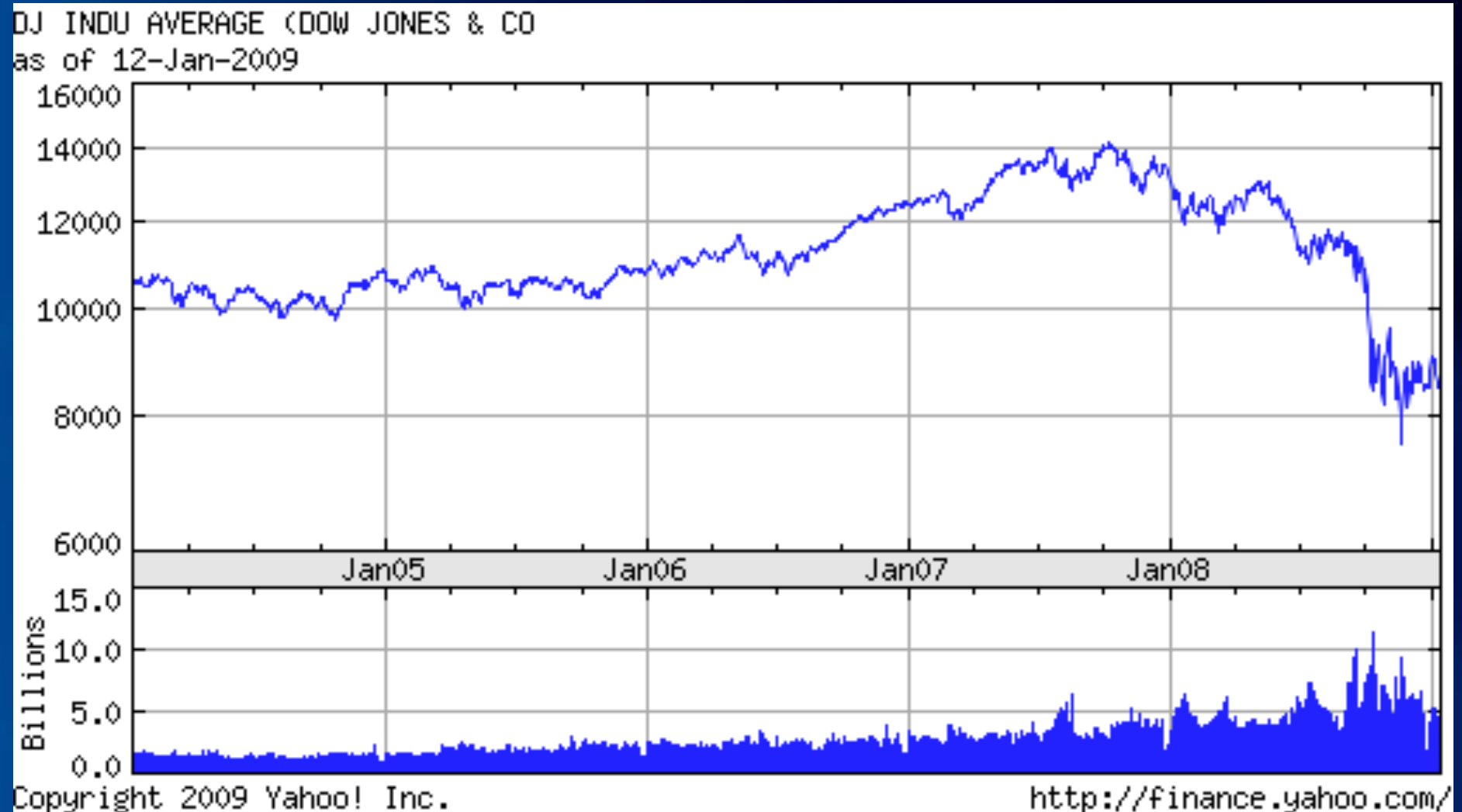
Types of Data: Shuttle Launch Accelerations



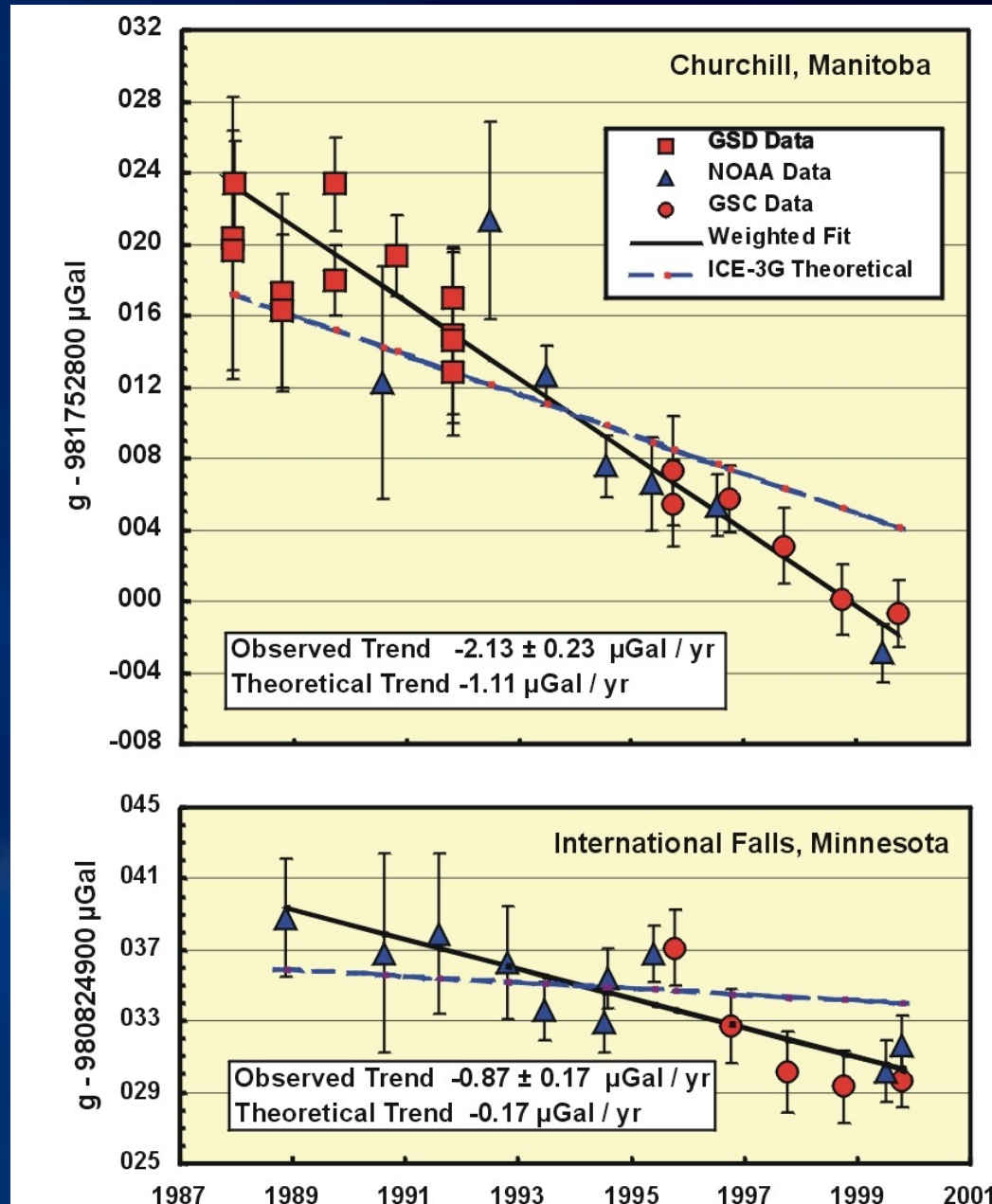
Types of Data: Election Analyses



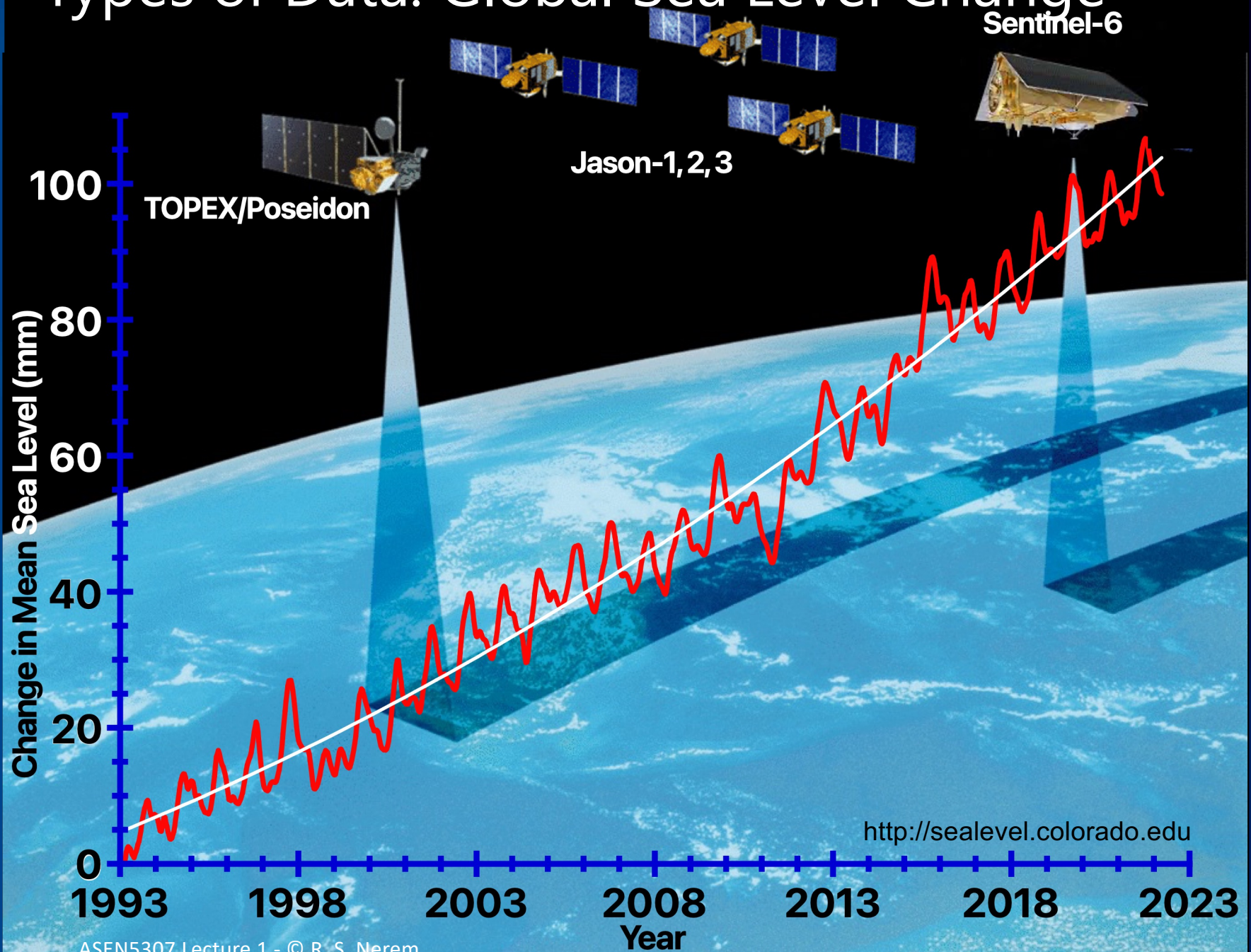
Types of Data: Dow Jones Industrial Avg



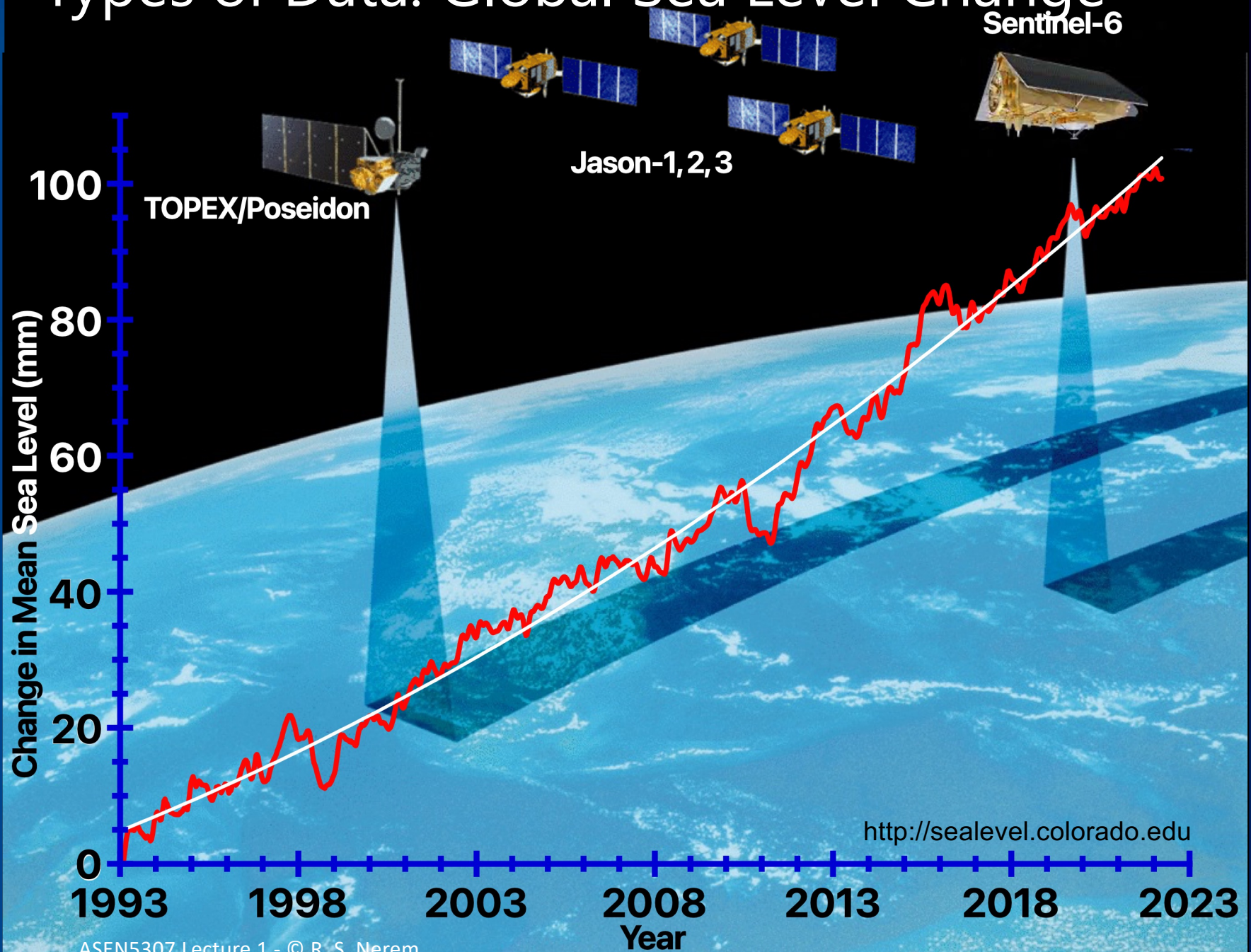
Types of Data: Gravity Measurements



Types of Data: Global Sea Level Change



Types of Data: Global Sea Level Change

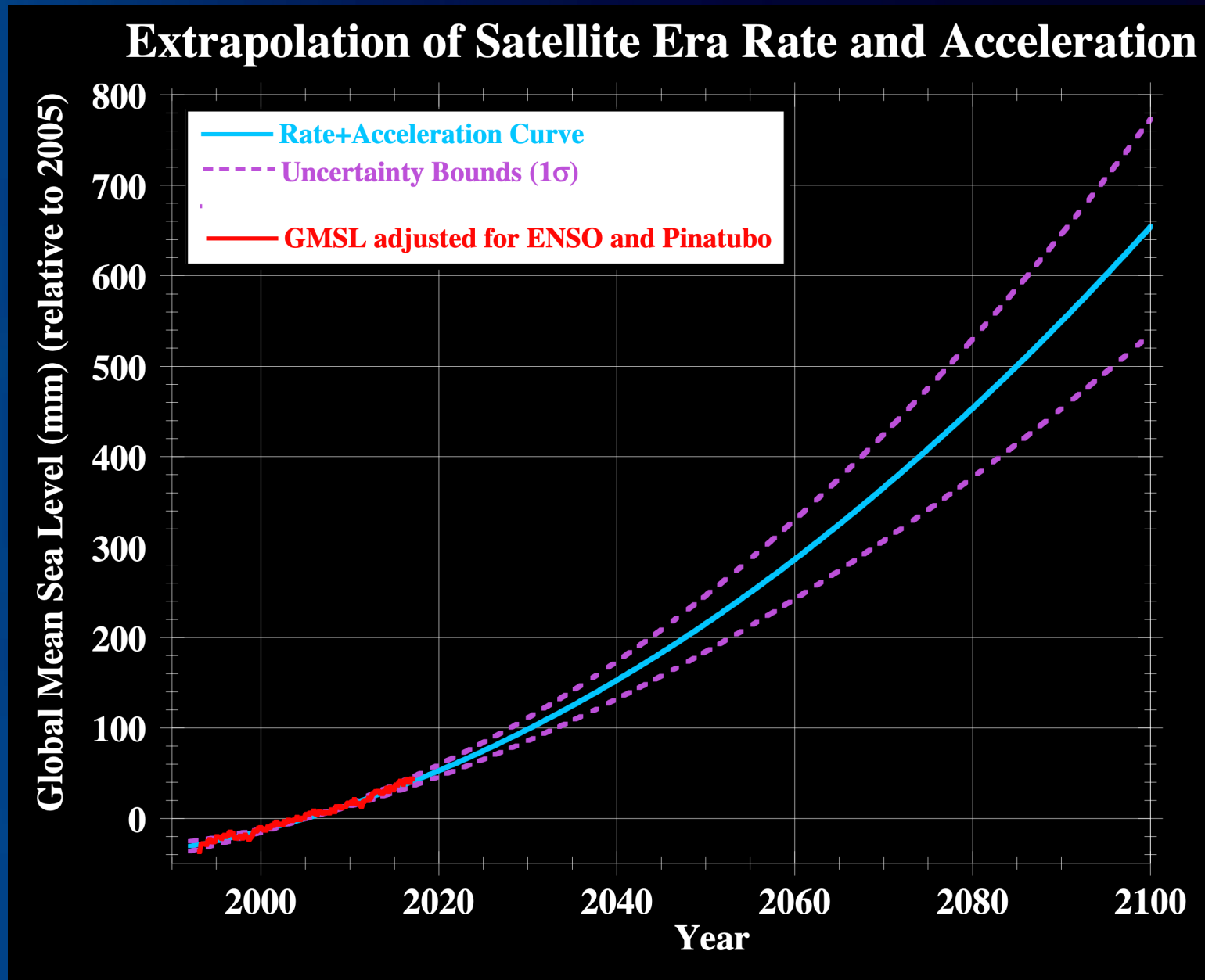


Types of Data: Global Sea Level Change

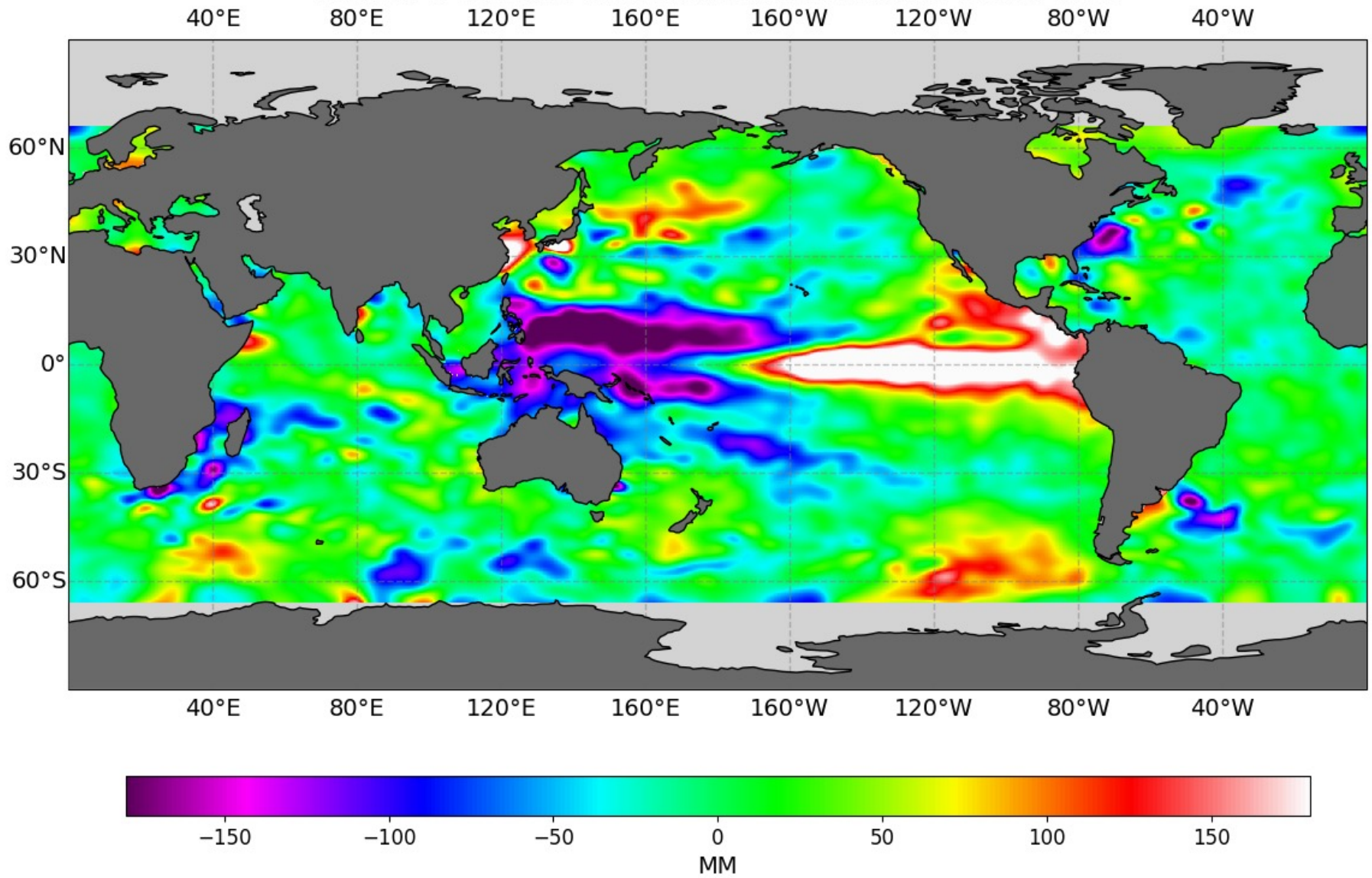
year	msl (mm)
1992.9595	-16.538
1992.9866	-20.771
1993.0138	-21.645
1993.0409	-20.067
1993.0681	-19.298
1993.0952	-23.830
1993.1223	-22.319
1993.1495	-20.311
1993.1766	-18.890
1993.2038	-13.430
1993.2309	-14.146

<http://sealevel.colorado.edu>

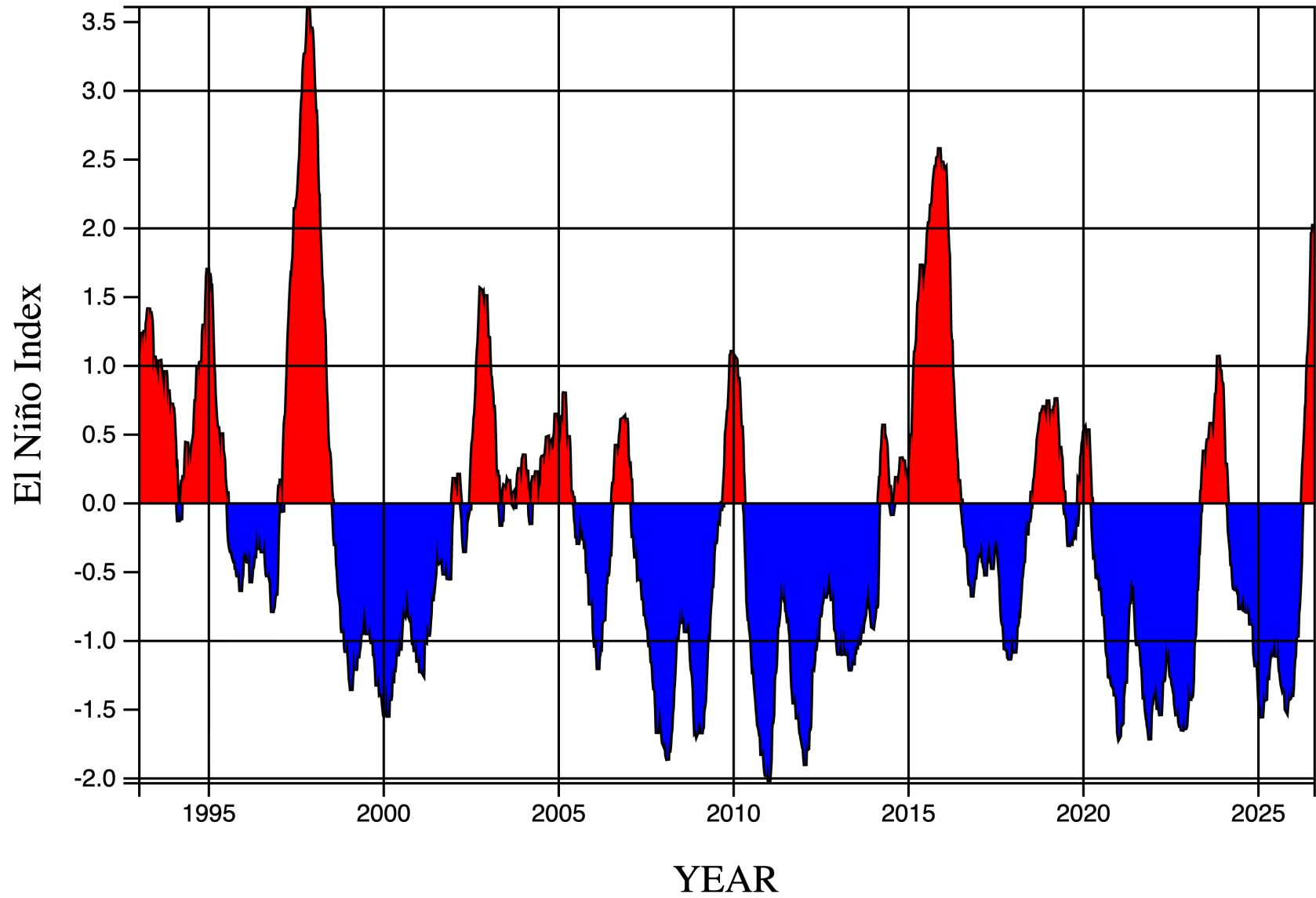
Types of Data: Global Sea Level Change



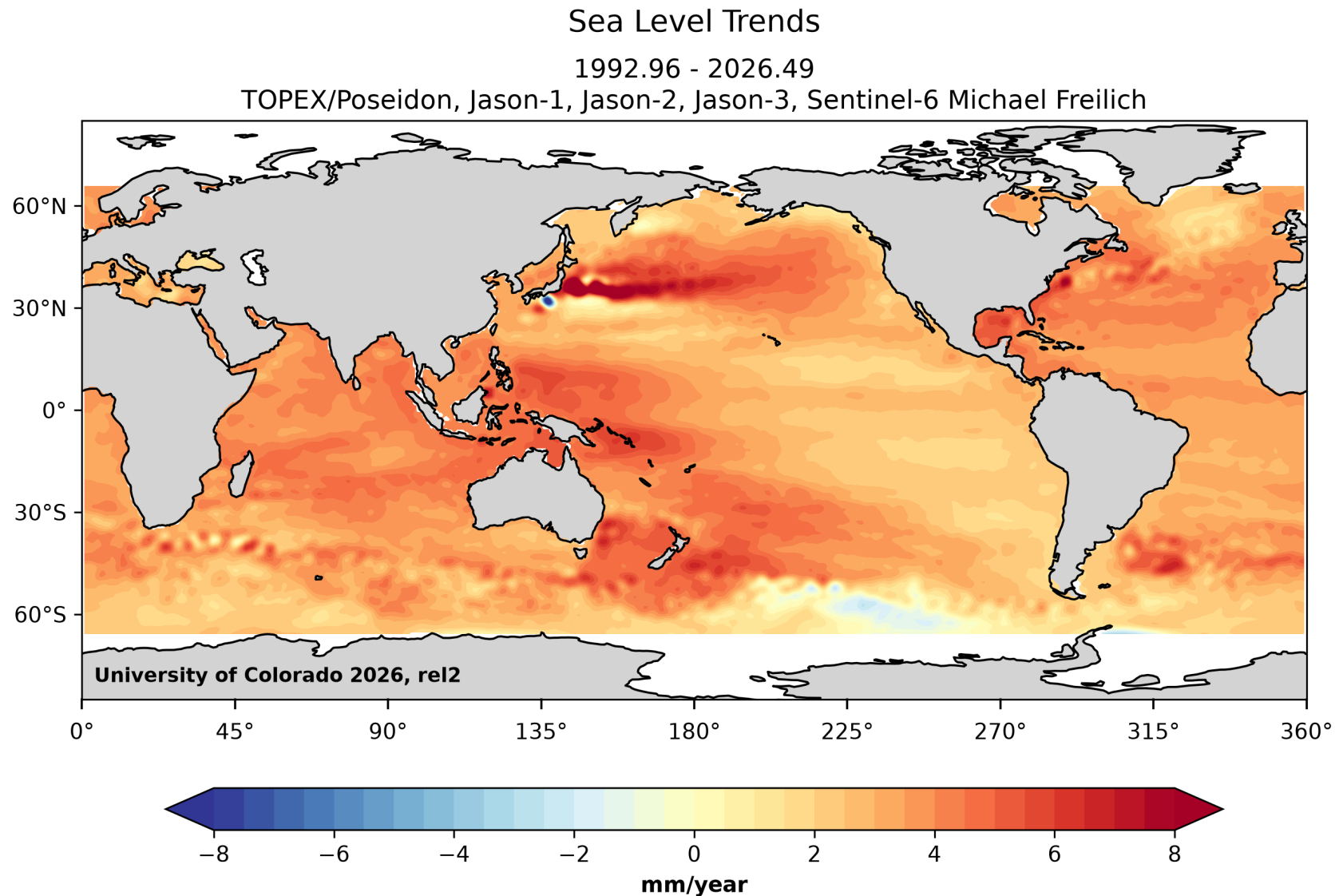
Sentinel-6 Michael Freilich Sea Level Residuals AUG 10 2026



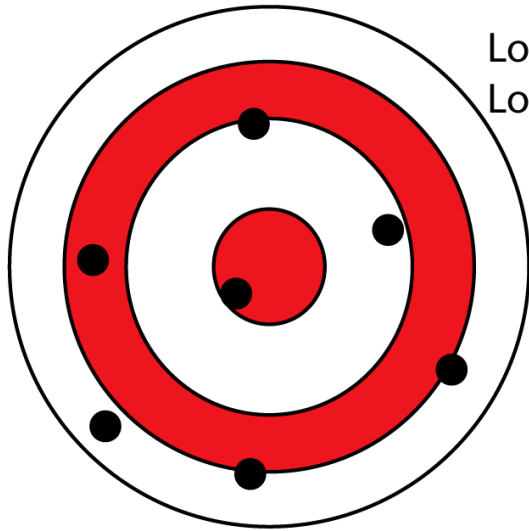
ENSO Index



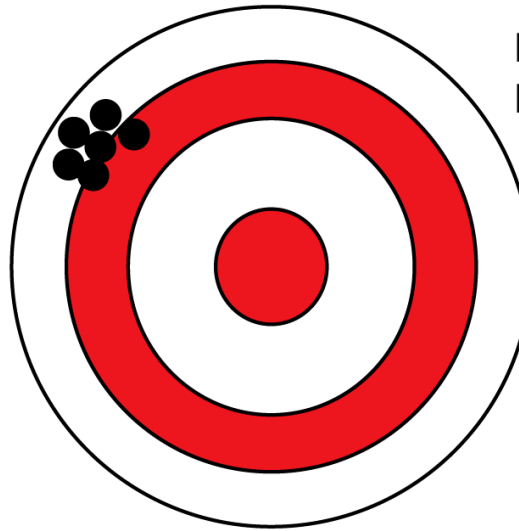
Types of Data: Global Sea Level Change



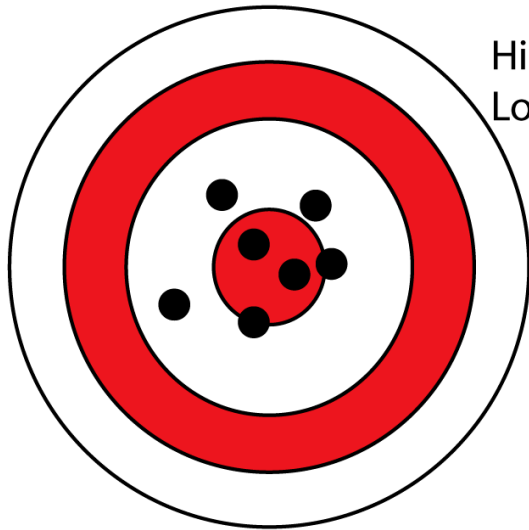
Accuracy and Precision



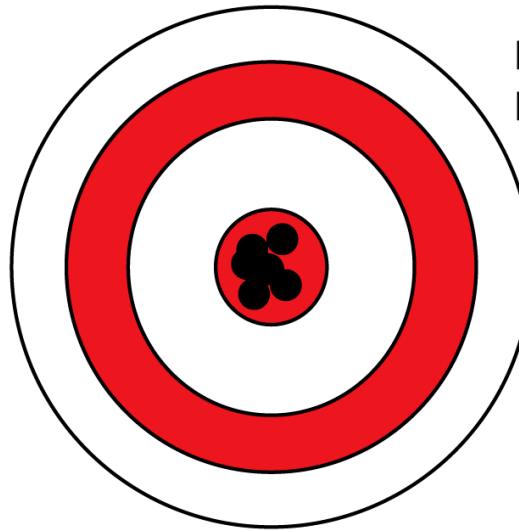
Low accuracy
Low precision



Low accuracy
High precision



High accuracy
Low precision



High accuracy
High precision

Accuracy and Precision

		Precision	
Accuracy	High	High	Low
	Low	49, 50, 50, 52, 50, 49, 51, 50	55, 47, 50, 52, 44, 53, 57, 47
		54, 55, 55, 57, 55, 54, 56, 55	60, 52, 44, 57, 49, 58, 62, 52
		Precision	
Accuracy	High	High	Low
	Low	50.1±1.0	50.6±4.4
		55.1±1.0	55.2±4.4

(Swan and Sandilands, 1996, Page 2, Box 1.1)

Types of Data

Ordinal Data



(<http://www.krantz-online.de>)

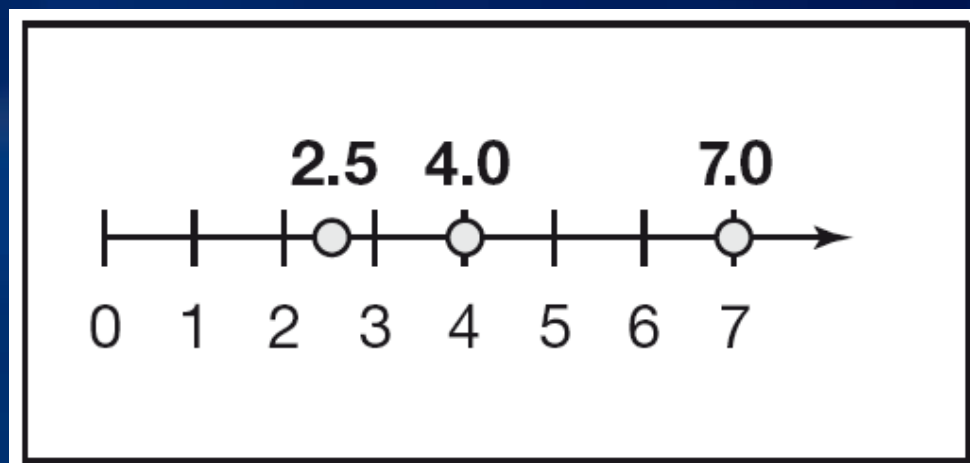
1. Talc
2. Gypsum
3. Calcite
4. Fluorite
5. Apatite
6. Orthoclase
7. Quartz
8. Topaz
9. Corundum
10. Diamond

(Trauth, 2007)

Ratio Data

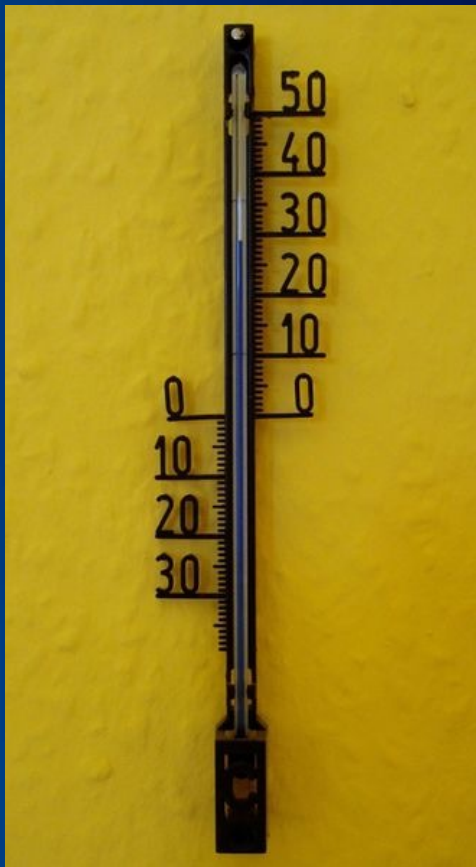


(Wikipedia Commons)

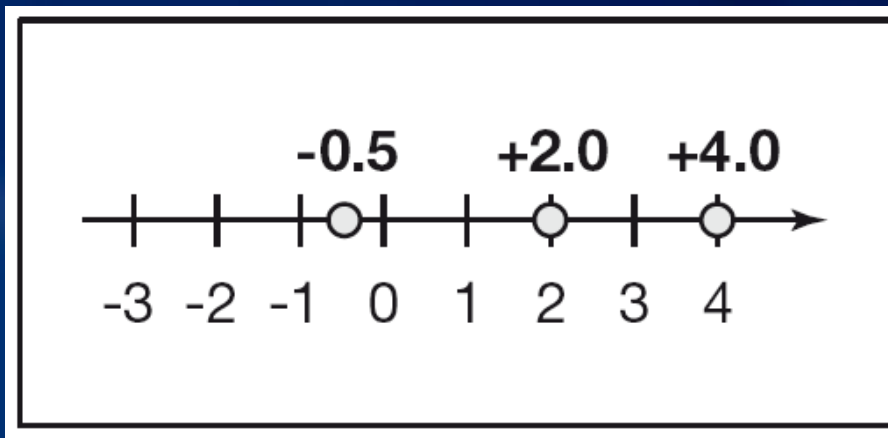


(Trauth, 2007)

Interval Data

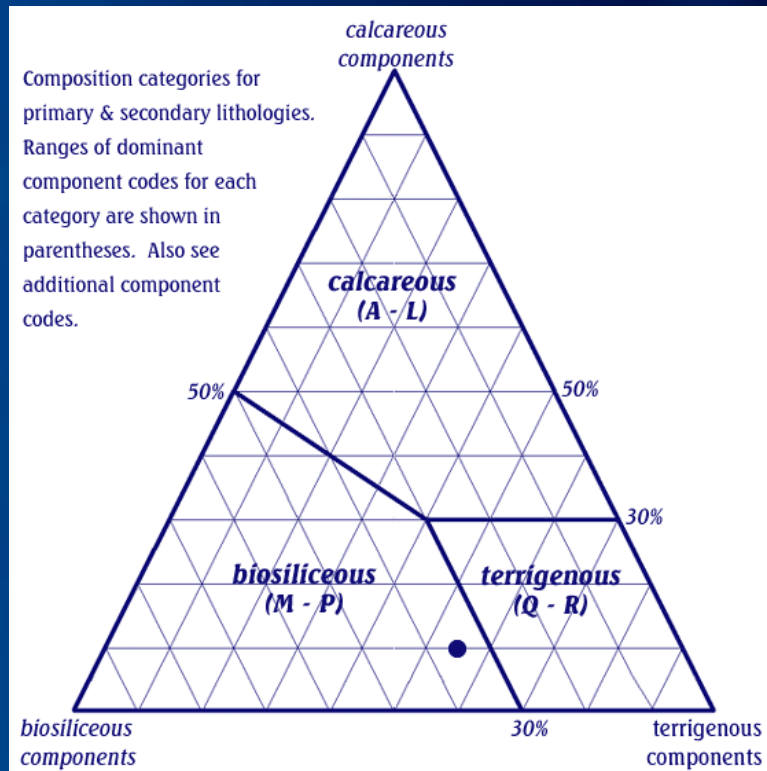


(Wikipedia Commons)

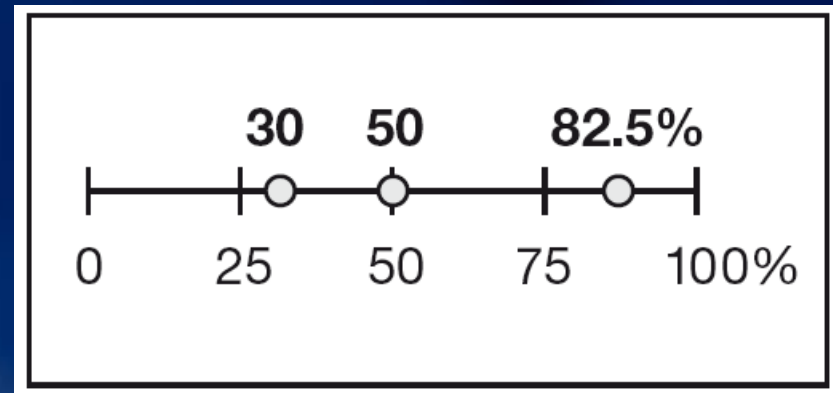


(Trauth, 2007)

Closed Data

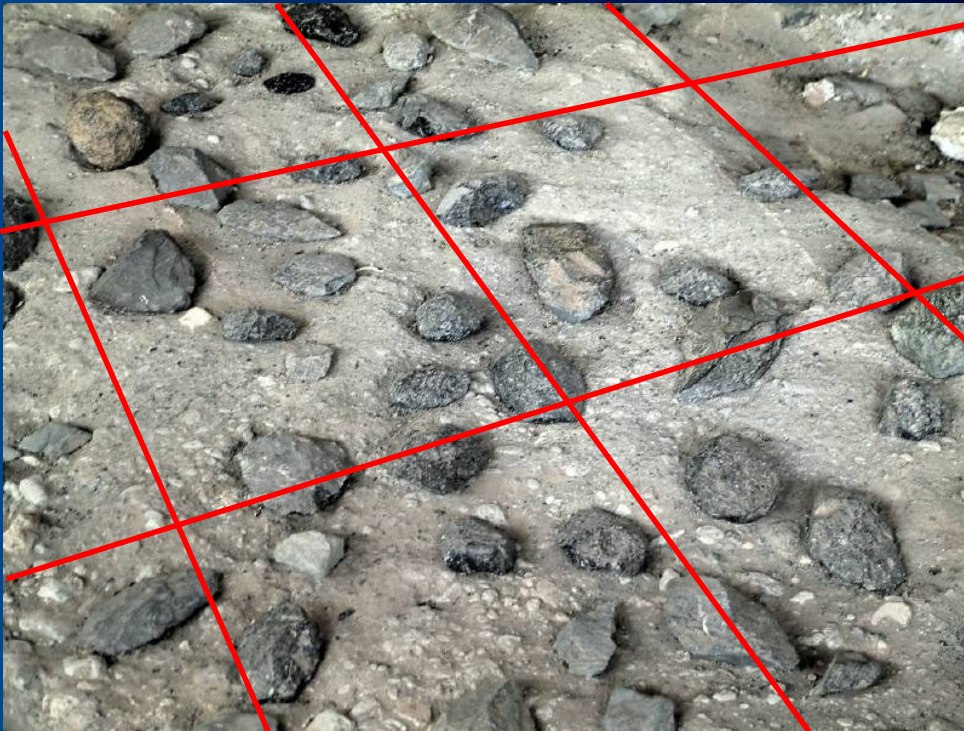


(<http://NOAA.gov>)

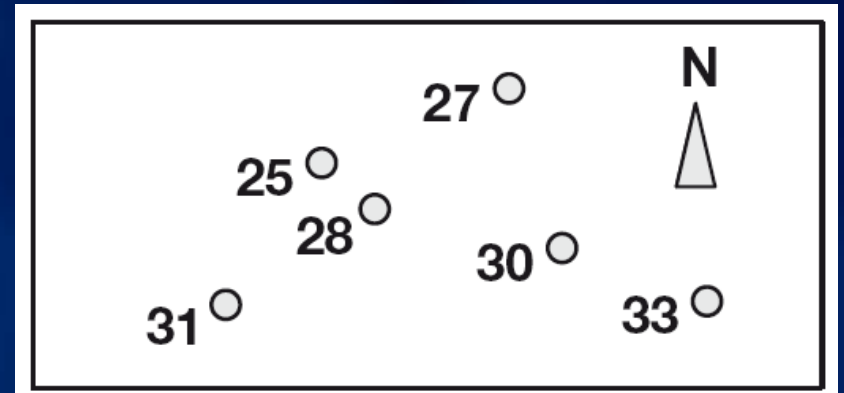


(Trauth, 2007)

Spatial Data



(Photo Trauth 2008)

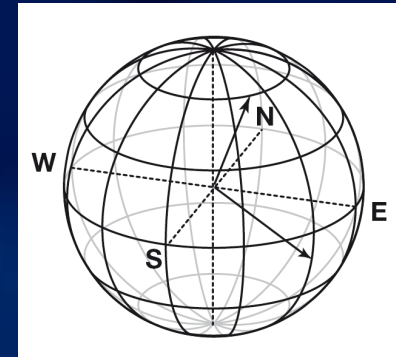
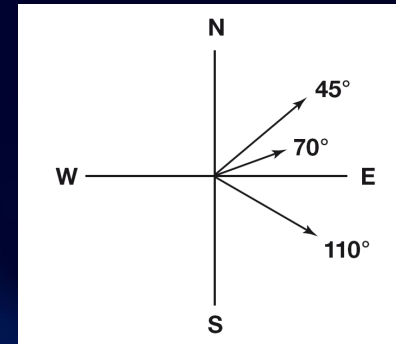


(Trauth, 2007)

Directional Data



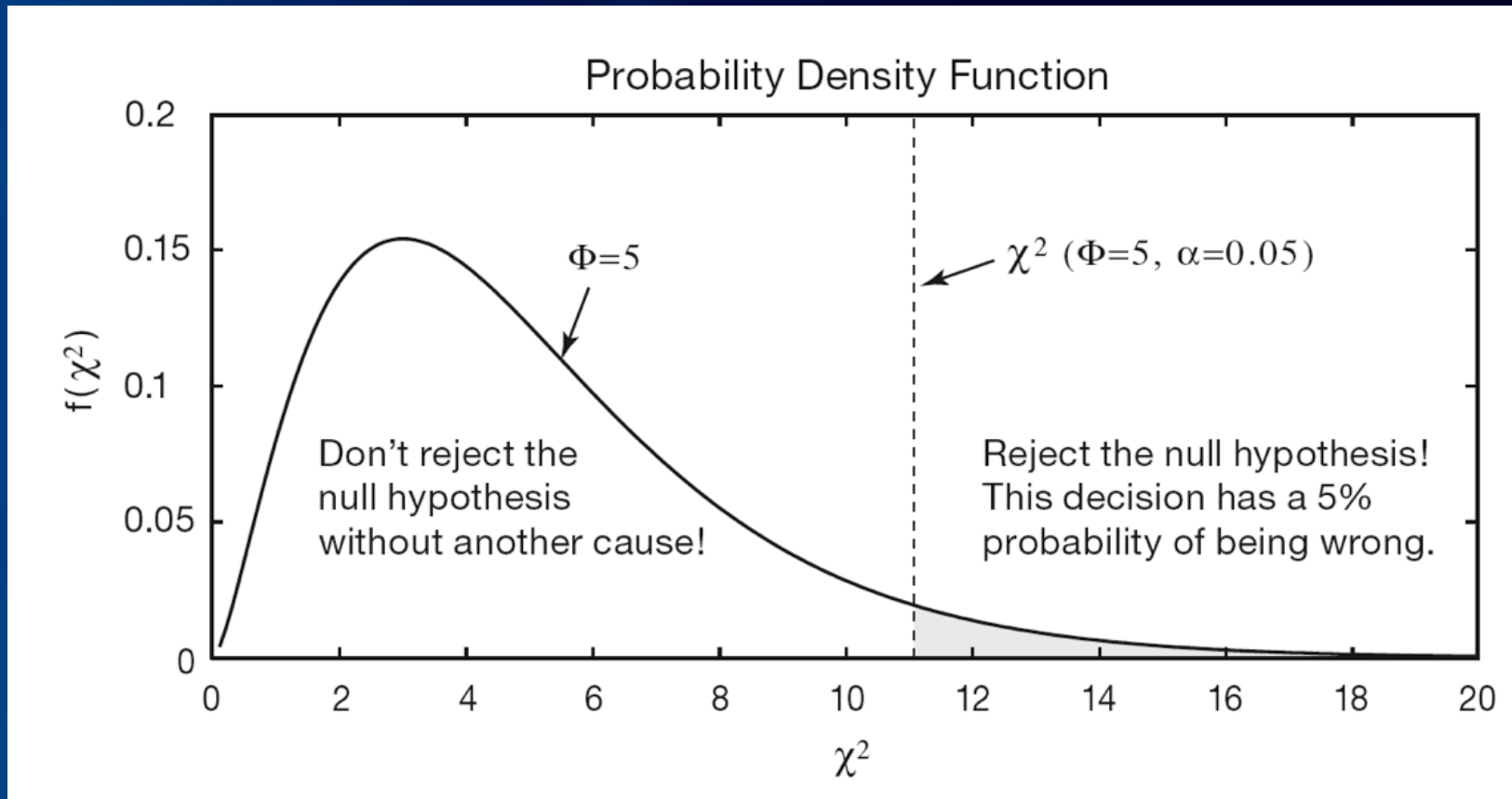
(Photo Trauth 2008)



(Trauth, 2007)

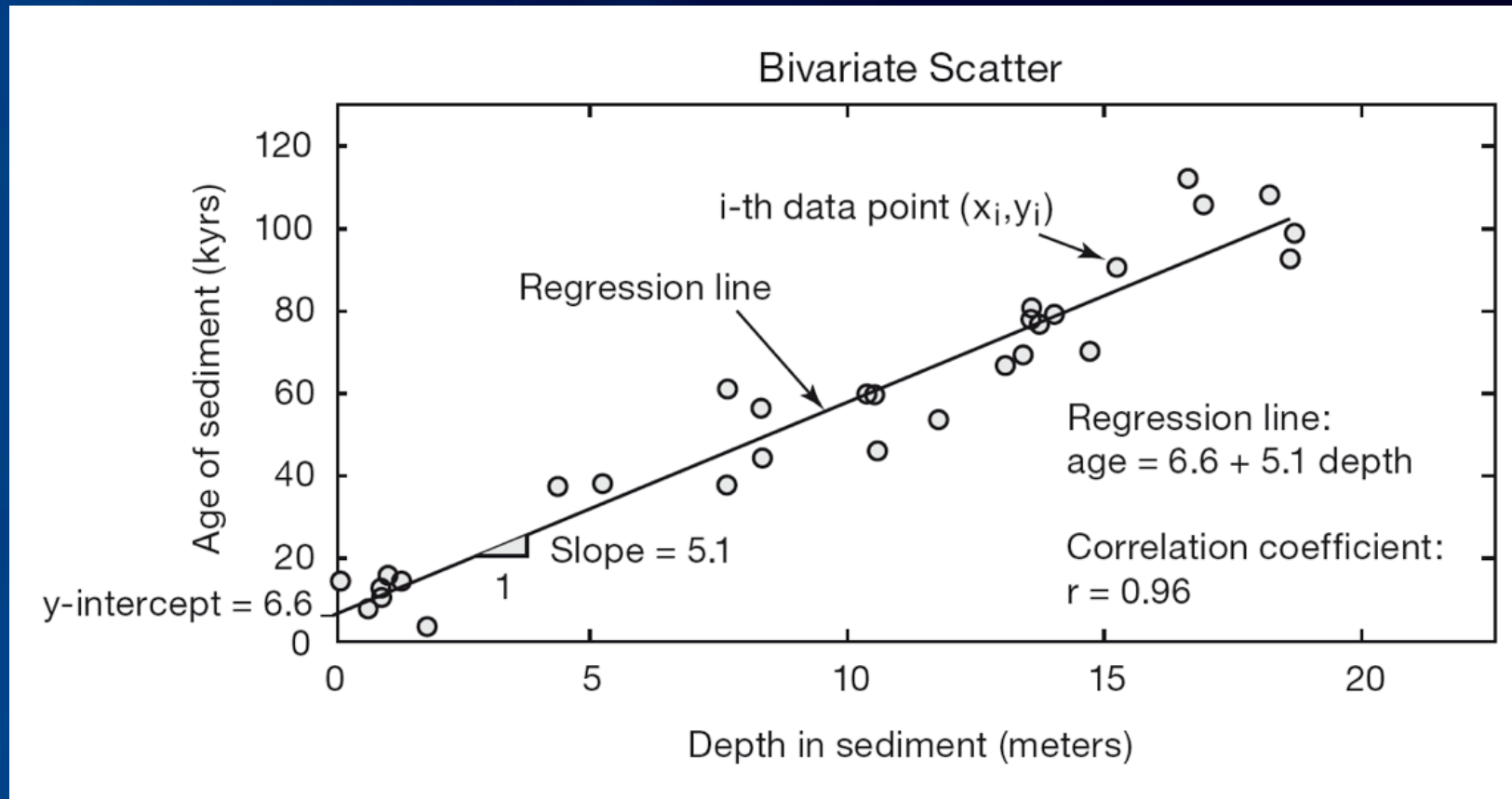
Methods

Univariate Statistics



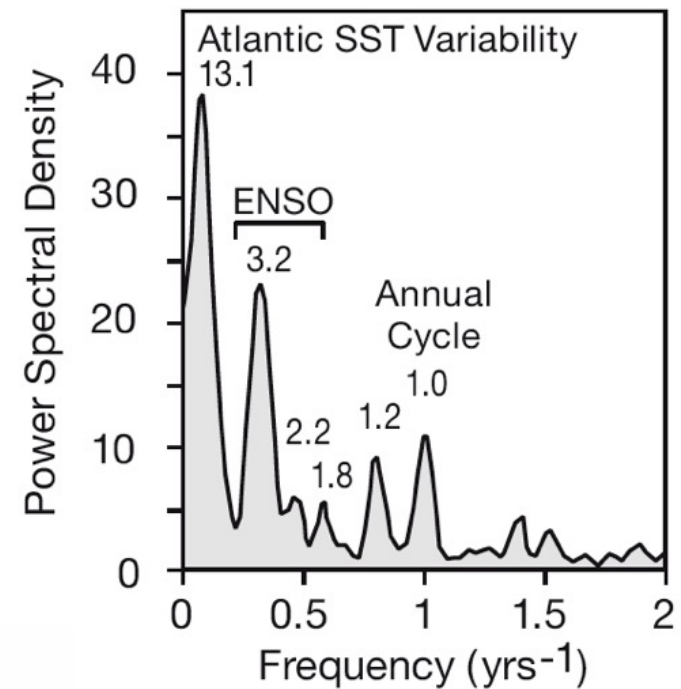
(Trauth, 2007)

Bivariate Statistics



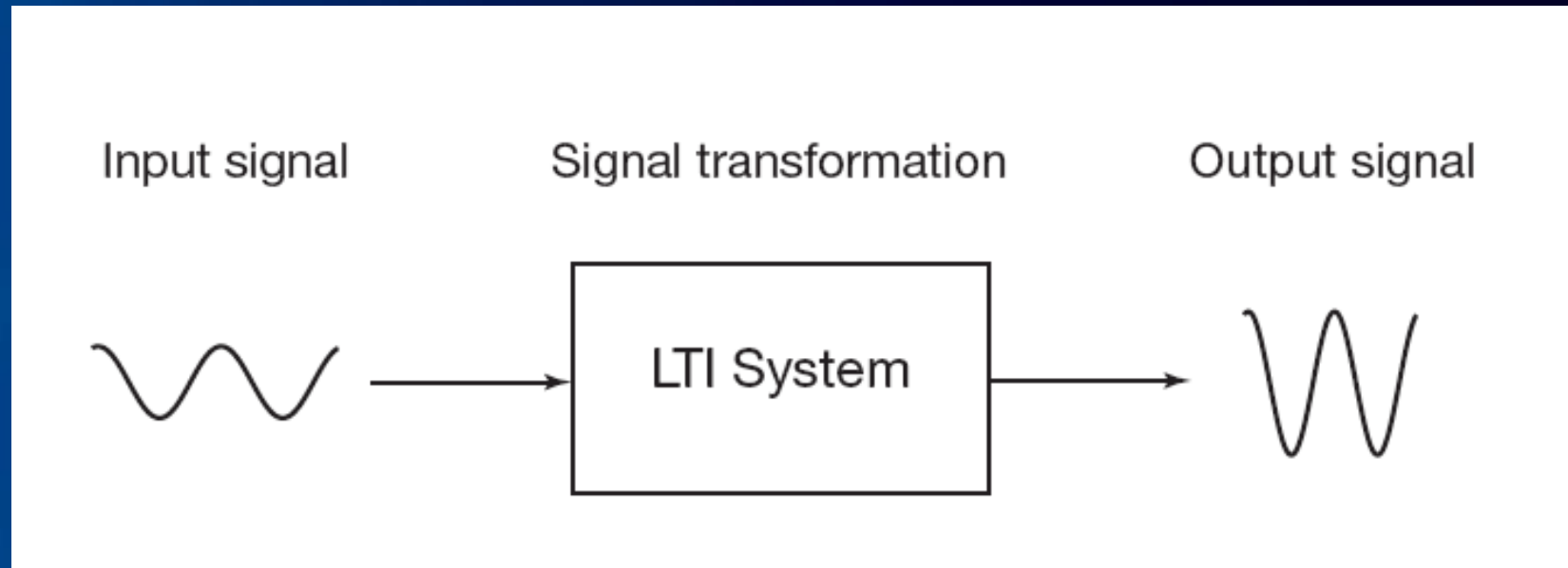
(Trauth, 2007)

Time-Series Analysis



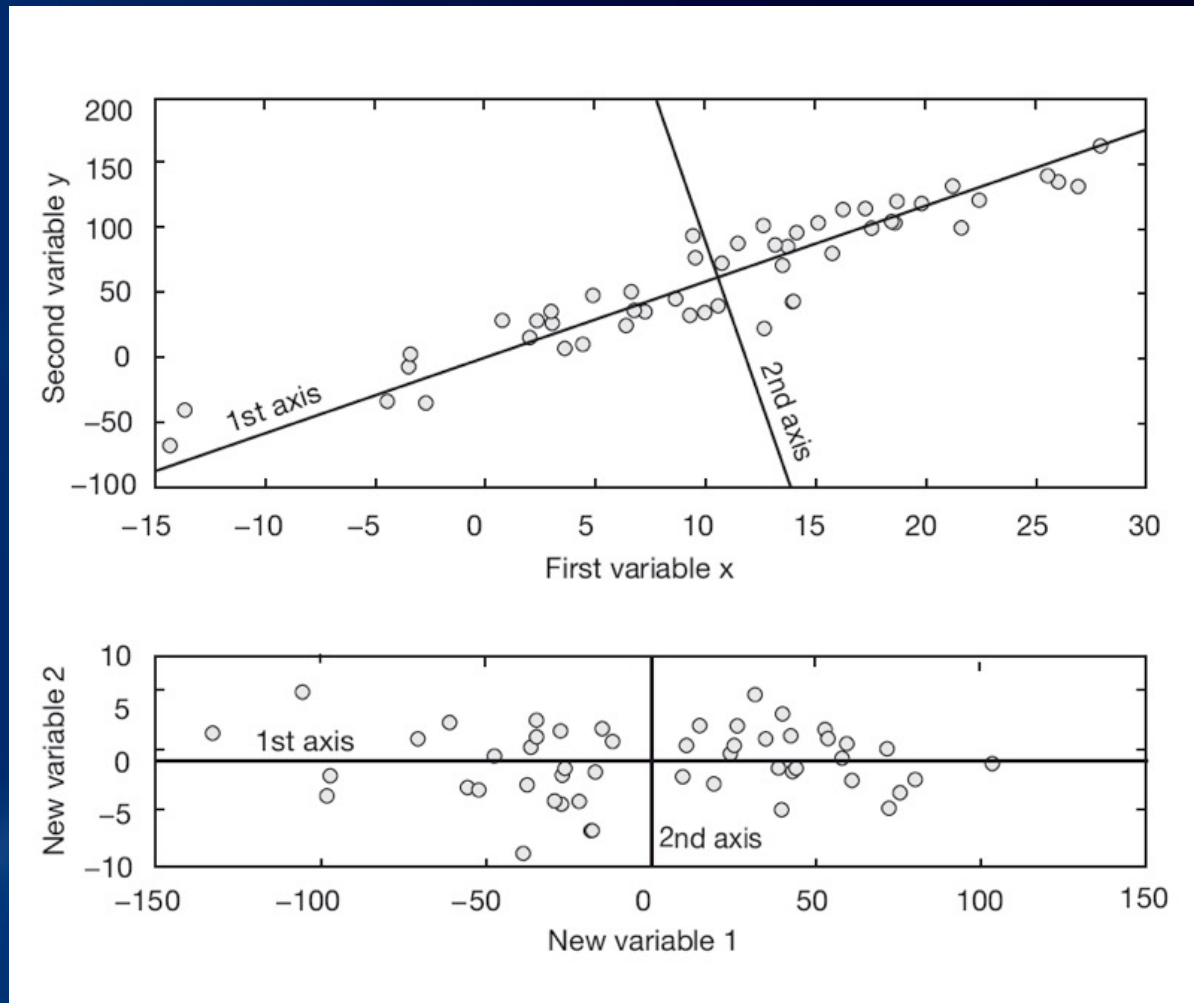
(Trauth, 2007)

Data Filtering



(Trauth, 2007)

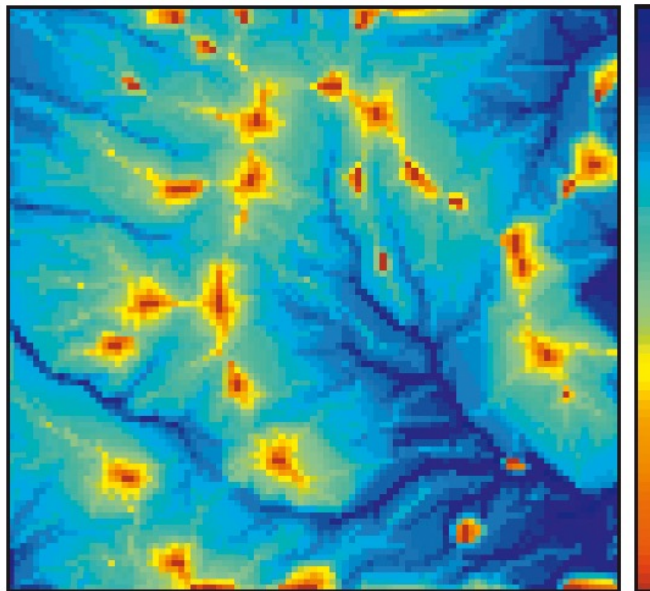
Principal Component Analysis



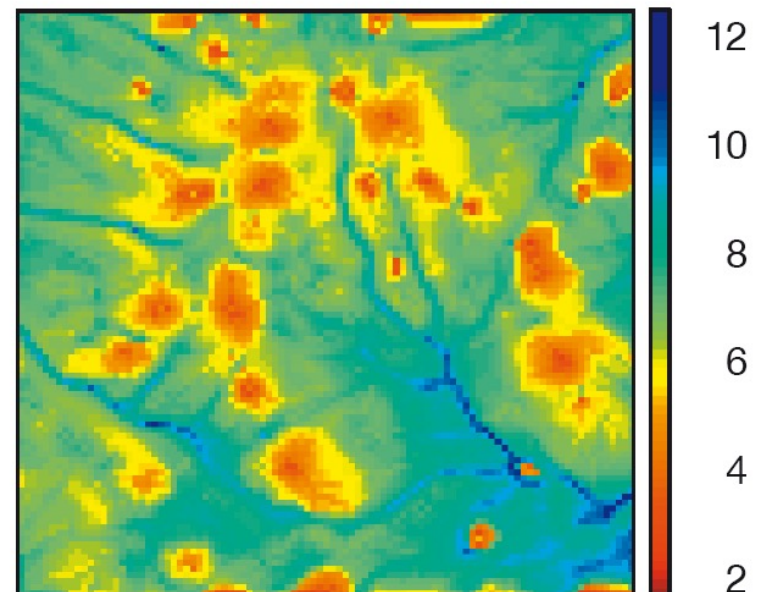
(Trauth, 2007)

Spatial Analysis

Flow Accumulation



Wetness Index



(Trauth, 2007)

Geostatistics

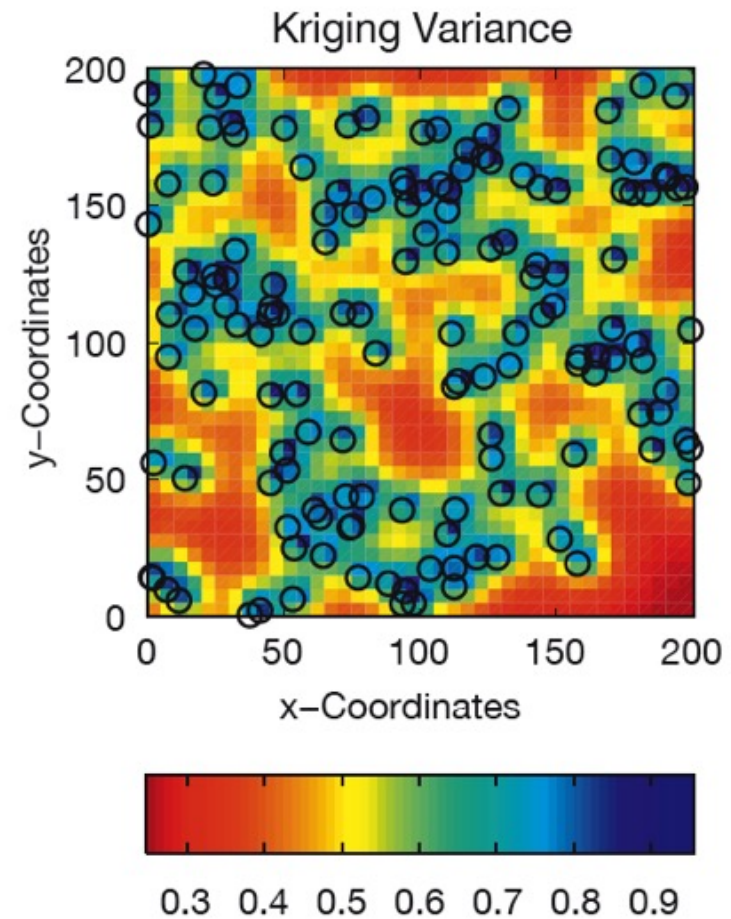
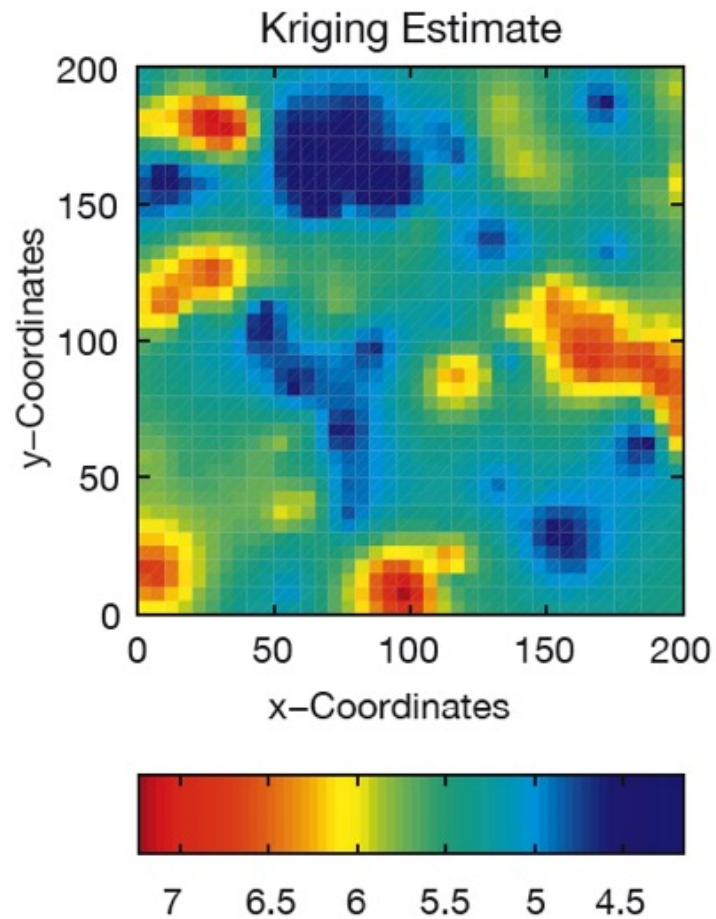
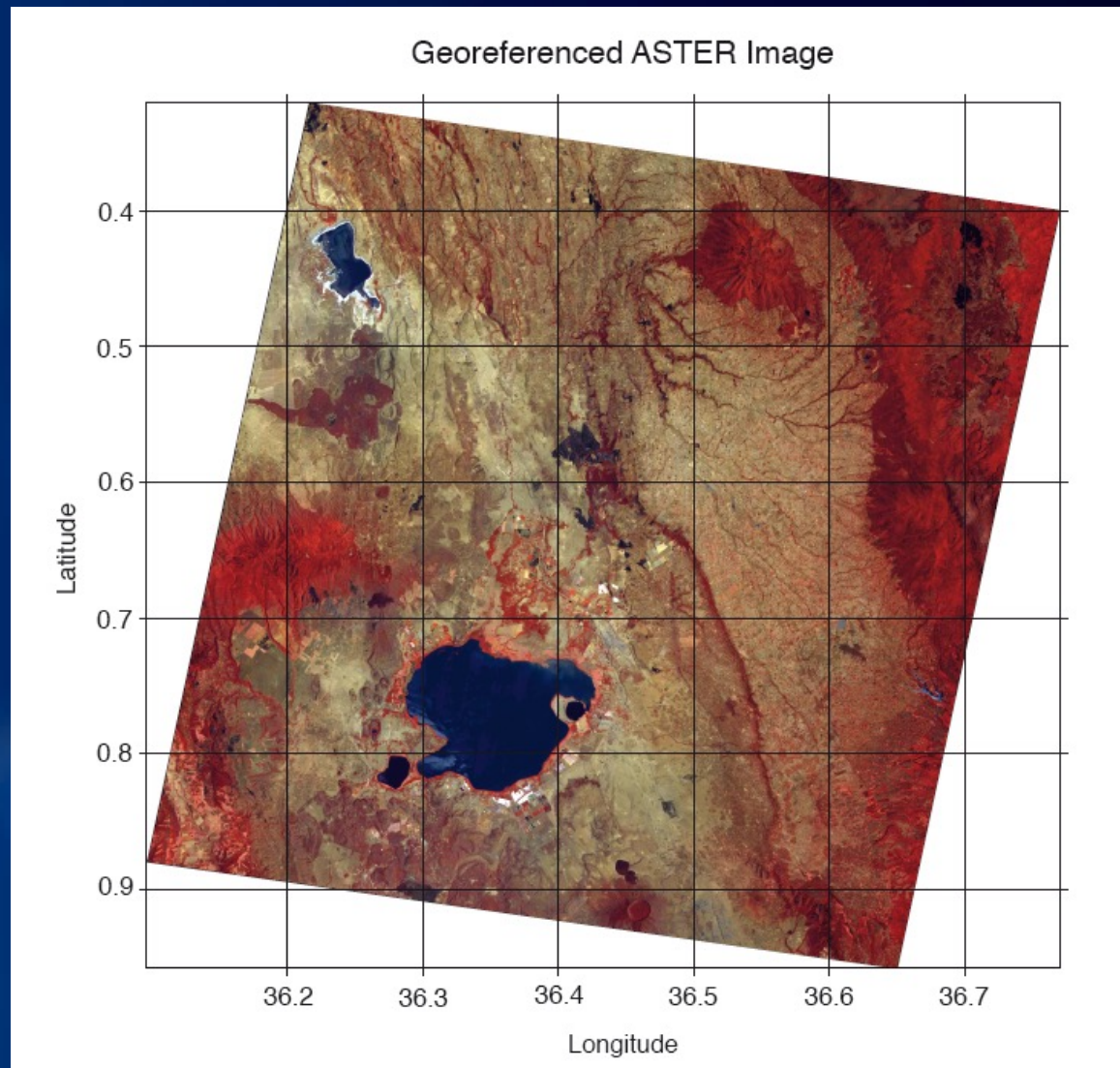
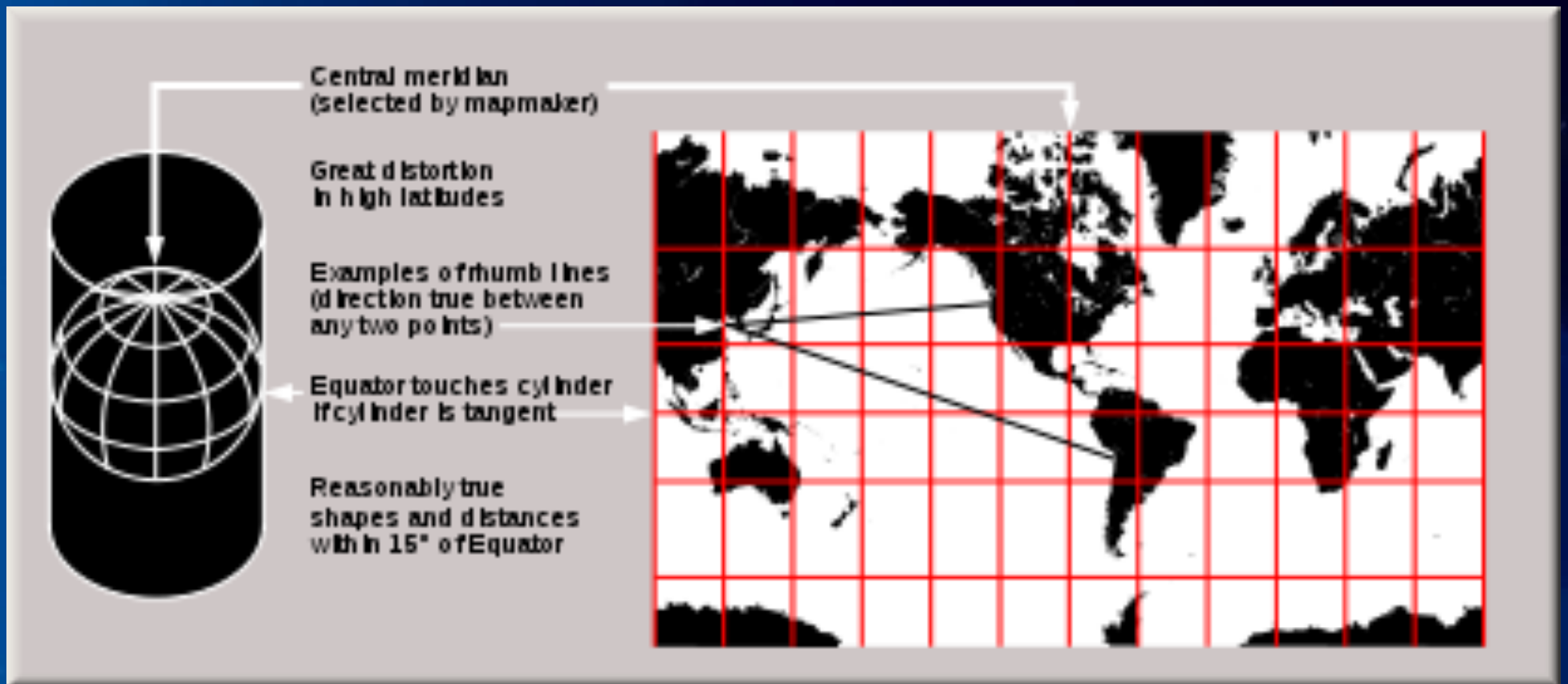


Image Analysis, Remote Sensing

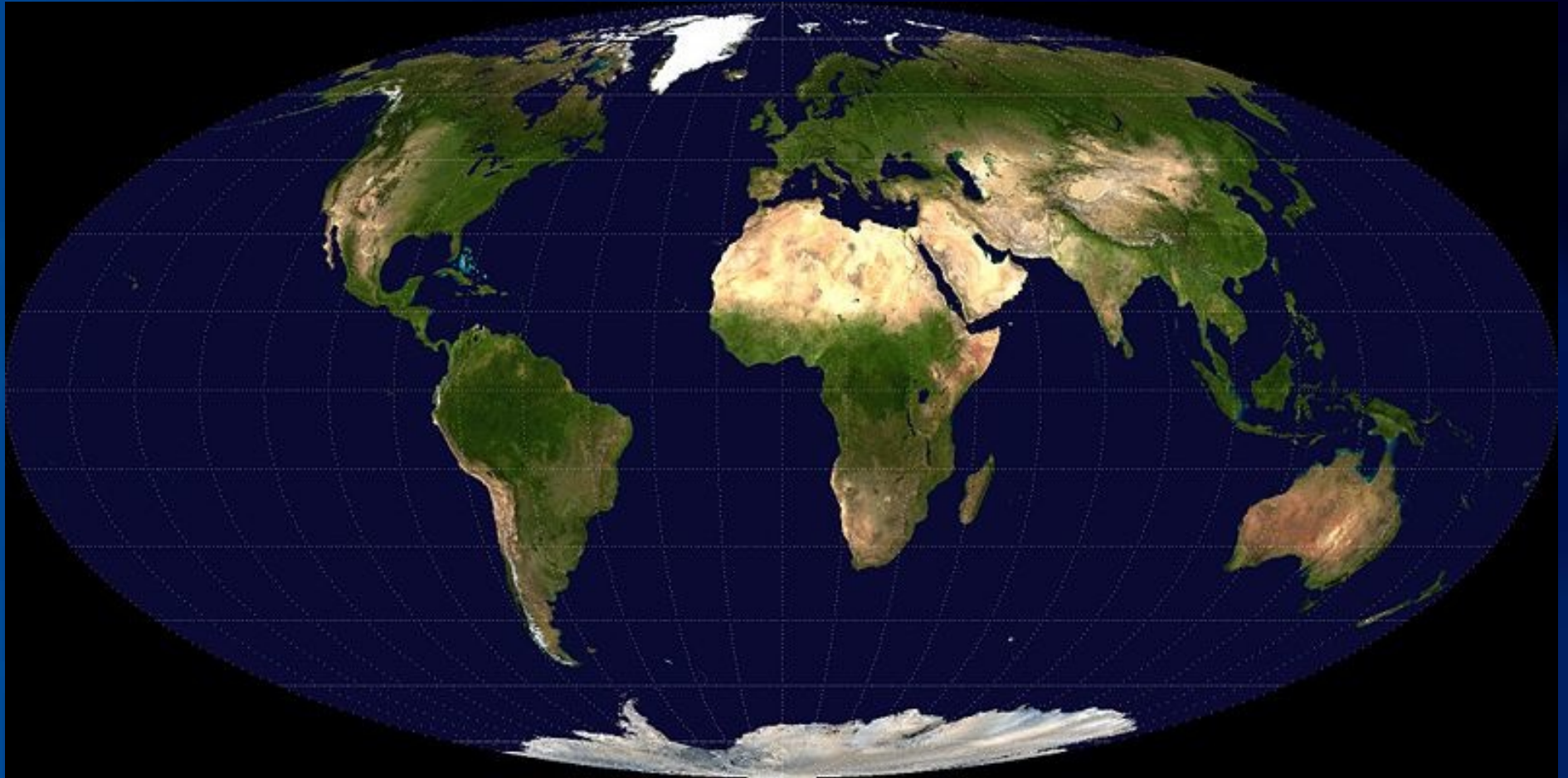


(Trauth, 2007 and <http://asterweb.jpl.nasa.gov/>)

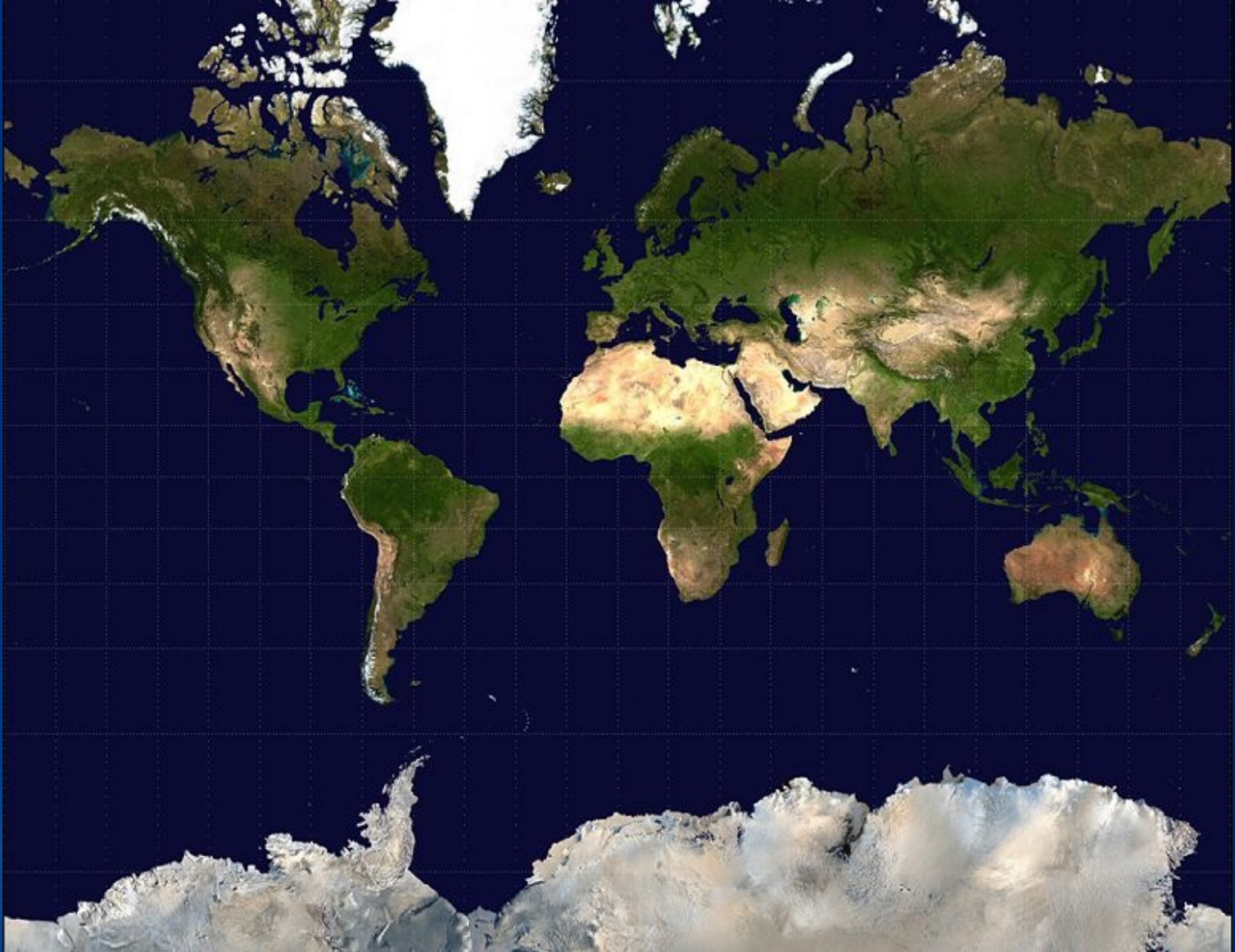
Map Projections



Mollweide Projection: Equi-Area

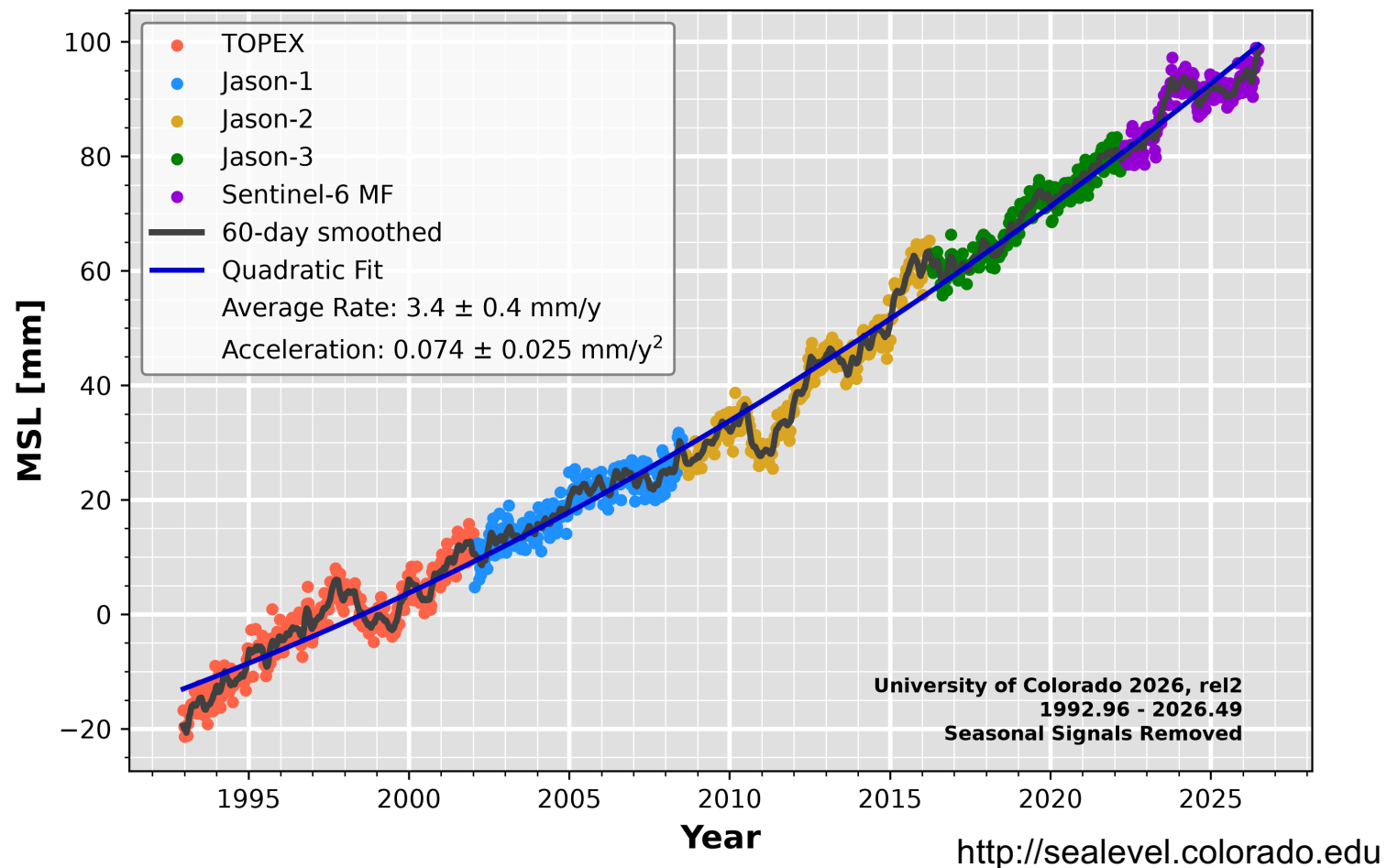


Mercator Projection: Cylindrical

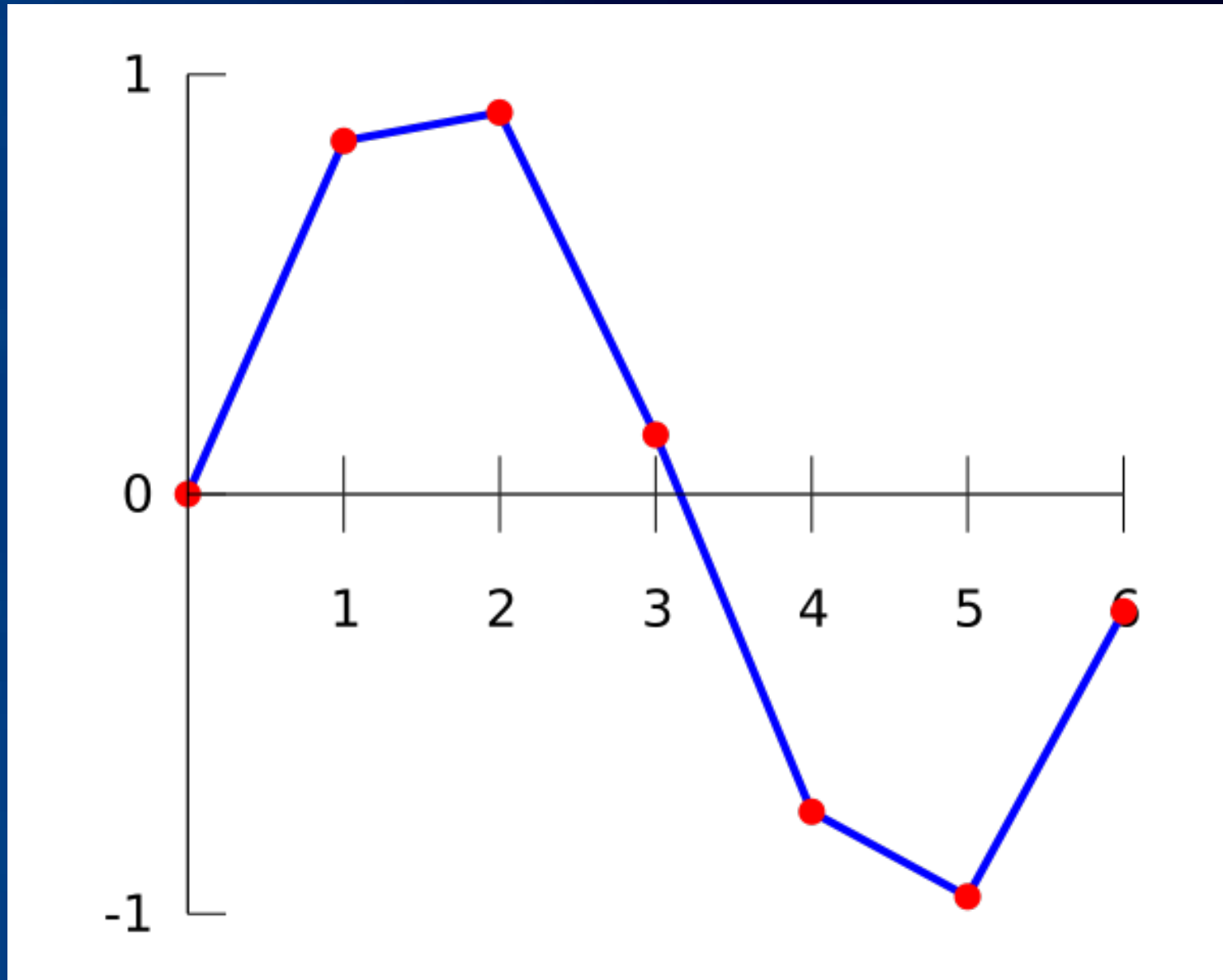


Data Calibration, Interpolation, Editing

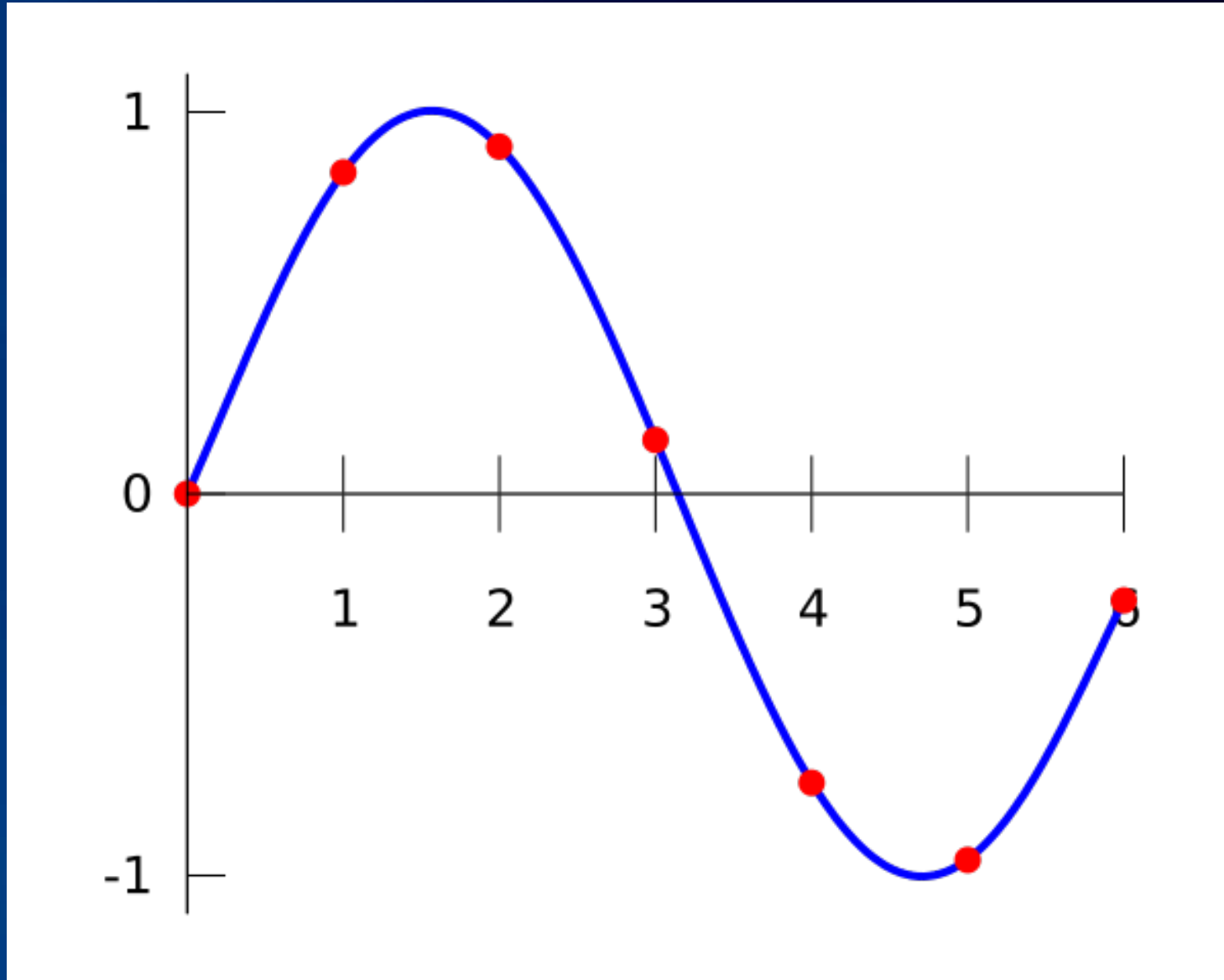
Data Calibration, Interpolation, Editing



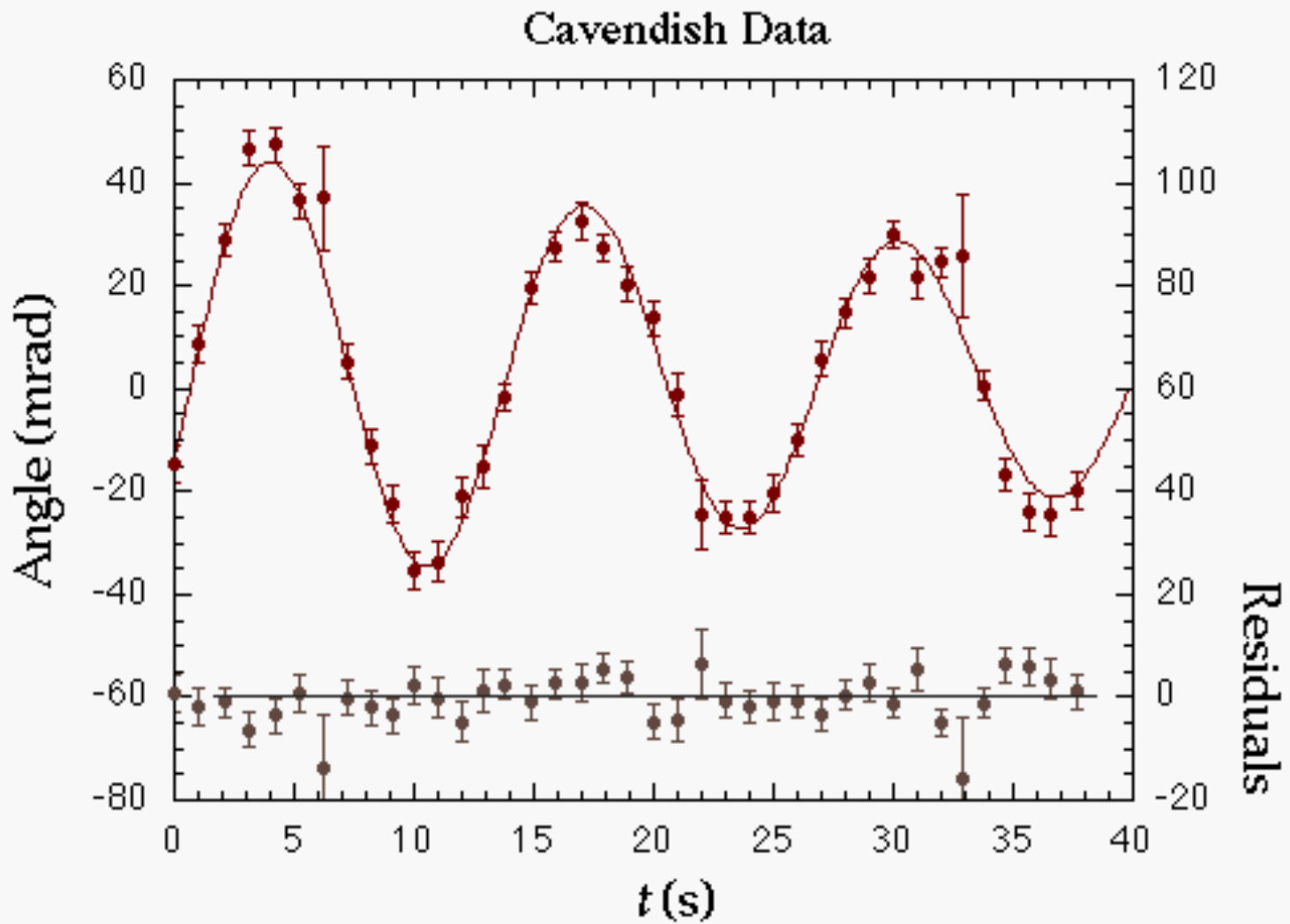
Data Interpolation: Linear Interpolation



Data Interpolation: Polynomial Interpolation



Data Editing



1 April 1798

CRIB SHEET 2.1 When first examining a dataset

Archive your data

Keep an unaltered copy of your data, together with notes recording its source and any other useful information that you come across.

Questions to ask yourself

What was the purpose for which the data was collected?

What is the physical significance of each data type?

How were they measured?

What are the units of measurement?

How are missing data represented?

Look at the numerical values

How many significant digits are recorded?

Make preliminary plots

time series plots

symbols or lines or both?

scatter plots

linear or logarithmic axes?

histograms

color images

plot error bars whenever possible!

Reality Checks

Does the data make sense in the context of what you already know?

sign

magnitude

minimum and maximum values

scale of variation with time and distance

daily, annual and other periodicities



End